Nederlandse norm

NEN-EN-IEC 61252

(en)

Elektro-akoestiek - Toestellen voor het meten van de blootstelling aan geluid van personen (IEC 61252:2025,IDT)

Electroacoustics - Personal sound exposure meters (IEC 61252:2025,IDT)

Vervangt NEN 11252:1995;
NEN-EN-IEC 61252:1995/A1:2001;
NEN-EN-IEC 61252:1995/A2:2017

ICS 13.140; 17.140.50

januari 2026

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- IEC 61252:2025,IDT

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN IEC 61252

November 2025

ICS 13.140; 17.140.50

Supersedes EN 61252:1995; EN 61252:1995/A1:2001; EN 61252:1995/A2:2017

English Version

Electroacoustics - Personal sound exposure meters (IEC 61252:2025)

Électroacoustique - Exposimètres acoustiques individuels (IEC 61252:2025)

Elektroakustik - Anforderungen an Personenschallexposimeter (IEC 61252:2025)

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European foreword

The text of document 29/1206/FDIS, future edition 2 of IEC 61252, prepared by TC 29 "Electroacoustics" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN IEC 61252:2025.

The following dates are fixed:

- latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement (dop) 2026-11-30
- latest date by which the national standards conflicting with the document have to be withdrawn (dow) 2028-11-30

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In the official version, for Bibliography, the following notes have to be added for the standard indicated:

IEC 61094-1	NOTE	Approved as EN 61094-1
IEC 61094-4	NOTE	Approved as EN 61094-4
IEC 61252:1993	NOTE	Approved as EN 61252:1995 (not modified)
ISO 9612	NOTE	Approved as EN ISO 9612

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Annex ZA
(normative)

Normative references to international publications with their corresponding European publications

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 Where an International Publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

NOTE 2 Up-to-date information on the latest versions of the European Standards listed in this annex is available here: www.cencenelec.eu.

Publication	Year	Title	EN/HD	Year
IEC 60942	-	Electroacoustics - Sound calibrators	EN IEC 60942	-
IEC 61000-4-3	2020	Electromagnetic compatibility (EMC) - Part 4-3 : Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test	EN IEC 61000-4-3	2020
IEC 61094-6	-	Measurement microphones - Part 6: Electrostatic actuators for determination of frequency response	EN 61094-6	-
IEC 61183	-	Electroacoustics - Random-incidence and diffuse-field calibration of sound level meters	EN 61183	-
IEC 61672	-	Electroacoustics-Sound level meters	-	-
IEC 61672-1	2013	Electroacoustics - Sound level meters - Part 1: Specifications	EN 61672-1	2013
IEC 61672-2	2013	Electroacoustics - Sound level meters - Part 2: Pattern evaluation tests	EN 61672-2	2013
IEC 61672-3	2013	Electroacoustics - Sound level meters - Part 3: Periodic tests	EN 61672-3	2013
IEC 62585	2012	Electroacoustics - Methods to determine corrections to obtain the free-field response of a sound level meter	EN 62585	2012
ISO/IEC Guide 98-3	-	Uncertainty of measurement - Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)	-	-
ISO/IEC Guide 98-4	-	Uncertainty of measurement -- Part 4: Role of measurement uncertainty in conformity assessment	-	-
ISO/IEC Guide 99	-	International vocabulary of metrology_- Basic and general concepts and associated terms (VIM)	-	-

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Electroacoustics - Personal sound exposure meters

FOREWORD

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IEC 61252 has been prepared by IEC technical committee 29: Electroacoustics. It is an International Standard.

This second edition cancels and replaces the first edition published in 1993, Amendment 1:2000, and Amendment 2:2017. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) personal sound exposure meters are required to provide indications of time-averaged sound level and peak sound level;

b) sound exposure is an optional quantity for indication;

c) specifications for physical quantities that do not follow the principle of equal-energy exchange rate have been added;

d) specifications for directional response have been added;

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e) specifications for frequency weightings apply to the relative diffuse-field frequency response;

f) determination of conformance to specifications takes account of uncertainties of measurement;

g) detailed requirements for pattern-evaluation tests and periodic testing have been added.

The text of this International Standard is based on the following documents:

Draft	Report on voting
29/1206/FDIS	29/1218/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with ISO/IEC Directives, Part 2, and developed in accordance with ISO/IEC Directives, Part 1 and ISO/IEC Directives, IEC Supplement, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications.

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INTRODUCTION

The principal application for a personal sound exposure meter is the measurement of sound immission in the vicinity of a person's head, for example for assessment of occupational noise exposure in accordance with International Standards such as ISO 1999 and ISO 9612.

The microphone of a personal sound exposure meter is typically worn on the shoulder, collar, or other location close to the ear. In many practical situations, the sound immission indicated by an instrument worn on a person is likely to be different from that which would be measured in the absence of a person. The influence of the person wearing a personal sound exposure meter should be considered when estimating the sound immission that would have been measured with the person absent.

The most common physical quantities for characterisation of sound immission are time-averaged sound levels and peak sound levels. For this reason, this document requires a personal sound exposure meter to provide indications of these sound levels.

Other quantities which are sometimes measured include sound exposure and sound exposure levels. Therefore, this document optionally allows a personal sound exposure meter to indicate these quantities.

Sound exposure is a physical measure that accounts for both the sound pressure and its duration, at a given location, through the time integral of the square of the instantaneous frequency-weighted sound pressure. A doubling (or halving) of the integration time of a constant sound level yields a doubling (or halving) of sound exposure. Similarly, an increase (or decrease) of 3 dB in a constant input sound level for a constant integration time yields a doubling (or halving) of the sound exposure. The same operating principle ("equal-energy exchange rate") underlies the measurement of sound exposure level.

The term "dose" is sometimes used to refer to a percentage of a predetermined criterion for exposure to noise in terms of a specified upper limit (often a legal limit) of a specified physical quantity. The physical quantity and the value of the limit vary between jurisdictions, and some jurisdictions specify quantities that are not based on the principle of equal-energy exchange. Therefore, this document allows a personal sound exposure meter to indicate these quantities and distinguishes between equal-energy-based and non-equal-energy-based quantities.

This document specifies performance requirements for personal sound exposure meters of one performance class. The specifications generally correspond to those for a class 2 integrating-averaging sound level meter as given in IEC 61672-1:2013 for an A-weighted sound pressure level range at least from 67 dB to 137 dB and a nominal frequency range from 20 Hz to 8 kHz. The design goals and the acceptance limits for deviations from the design goals are representative of the performance of practical instruments. Personal sound exposure meters are unlikely to be suitable for measurement of sound levels outside these ranges.

The title of this document has been changed with respect to IEC 61252:1993 because this document includes requirements for pattern-evaluation tests and periodic testing in addition to performance specifications.

The purpose of pattern evaluation is to determine whether a model of personal sound exposure meter conforms to all the performance specifications given in this document.

The purpose of periodic testing is to assure the user that the individual personal sound exposure meter conforms to the applicable performance specifications for a limited set of key tests and for the environmental conditions under which the tests are performed. The extent of the periodic tests is deliberately restricted to the minimum considered necessary. Because of the limited extent of the periodic tests, evidence of pattern approval is necessary to state that the individual personal sound exposure meter conforms to the complete set of specifications of this document.

The aim is to ensure that pattern evaluation and periodic testing are performed in a consistent manner by all laboratories.

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1 Scope

This document specifies

– performance specifications for personal sound exposure meters,
– details of the tests necessary to verify conformance to all mandatory specifications for the purpose of pattern evaluation, and
– procedures for periodic testing of a personal sound exposure meter.

Personal sound exposure meters conforming to the requirements of this document have a specified frequency response for sound incident on the microphone from all directions.

This document is applicable to instruments that are designed to be worn on a person in a configuration specified by the manufacturer for the measurement of sound immission resulting from steady, intermittent, fluctuating, irregular, or impulsive sounds. For reproducibility of results, specifications and tests for the response to sound waves apply without an operator present in the sound field.

Pattern evaluation tests and periodic tests described in this document apply to personal sound exposure meters for which the manufacturer claims conformance to the specifications given in this document.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60942, *Electroacoustics - Sound calibrators*

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IEC 61672 (all parts), *Electroacoustics - Sound level meters*

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IEC 61672-2:2013, *Electroacoustics - Sound level meters - Part 2: Pattern evaluation tests*

IEC 61672-3:2013, *Electroacoustics - Sound level meters - Part 3: Periodic tests*

IEC 62585:2012, *Electroacoustics - Methods to determine corrections to obtain the free-field response of a sound level meter*

ISO/IEC Guide 98-3, *Uncertainty of measurement - Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*

ISO/IEC Guide 98-4, *Uncertainty of measurement - Part 4: Role of measurement uncertainty in conformity assessment*

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ISO/IEC Guide 99, *International vocabulary of metrology - Basic and general concepts and associated terms (VIM)*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61000-4-3, IEC 61183, IEC 61672 (all parts), IEC 62585, ISO/IEC Guide 98-3, ISO/IEC Guide 98-4, ISO/IEC Guide 99, and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at https://www.electropedia.org/
- ISO Online browsing platform: available at https://www.iso.org/obp

3.1
sound exposure
time integral of the square of a frequency-weighted sound pressure signal over a stated time interval or event of stated duration

Note 1 to entry: Duration of integration is included implicitly in the time integral and is not always reported explicitly, although it is useful to state the nature of the event. For measurements of sound exposure over a specified time interval, duration of integration is usually reported and indicated by a suitable subscript to the letter symbol, for example as $E_{\mathrm{A,1\,h}}$.

Note 2 to entry: In symbols and as an example, A-weighted sound exposure $E_{\mathrm{A,T}}$ is represented by Formula (1):

$$E_{\mathrm{A},T} = \int_{t_1}^{t_2} p_{\mathrm{A}}^2(t)\mathrm{d}t \tag{1}$$

where $p_{\mathrm{A}}^2(t)$ is the square of the A-weighted sound-pressure signal during integration time T starting at t_1 and ending at t_2.

Note 3 to entry: The unit of sound exposure is pascal-squared seconds ($\mathrm{Pa}^2\mathrm{s}$) if sound pressure is in pascals and running time is in seconds.

Note 4 to entry: For applications such as measurement of exposure to noise in the workplace, sound exposure in pascal-squared hours is more convenient than pascal-squared seconds.

[SOURCE: IEC 61672-1:2013, 3.11]

3.2
normalized 8 h-average sound level
level of the time-mean-square, A-weighted sound pressure during a normalization time period T_{n} of 8 h such that the sound exposure therefrom is equal to that of a time-varying sound at a place where total sound exposure occurs within a time period not necessarily 8 h

Note 1 to entry: The unit of normalized 8 h-average sound level is decibels.

Note 2 to entry: In symbols, a normalized 8 h-average sound level (letter symbol $L_{\mathrm{Aeq,8\,hn}}$), relative to the reference sound pressure p_0 and the 8 h normalization time period T_{n}, is related to sound exposure E in pascal-squared hours by Formula (2):

$$L_{\mathrm{Aeq,8\,hn}} = 10\lg\left[E\Big/\left(p_0^2 T_{\mathrm{n}}\right)\right] \tag{2}$$

or, alternatively, by Formula (3):

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$$E = \left(p_0^2 T_\mathrm{n}\right)\left[10^{\left(0{,}1\times L_\mathrm{Aeq,8\,hn}\right)}\right] \tag{3}$$

Note 3 to entry: For computations, a simpler form of Formula (2) for normalized 8 h-average sound level is obtained, for sound exposure E in pascal-squared hours, after substituting the values of 20 µPa for p_0 and 8 h for T_n, as in Formula (4):

$$L_\mathrm{Aeq,8\,hn} = 10\lg\left[\left(E\times 10^9\right)\big/3{,}2\right] \tag{4}$$

Note 4 to entry: When a total sound exposure is described indirectly by an A-weighted time-averaged sound pressure level $L_\mathrm{Aeq,T}$, for an averaging time T greater or less than the normalization time period of 8 h, normalized 8 h-average sound level can be determined from Formula (5):

$$L_\mathrm{Aeq,8\,hn} = L_{\mathrm{Aeq},T} + 10\lg\left(T/T_\mathrm{n}\right) \tag{5}$$

Note 5 to entry: Annex A provides a table of normalized 8 h-average sound levels and corresponding sound exposures. For example, a sound exposure of 1 $\mathrm{Pa^2h}$ (irrespective of the period of time over which it is measured) corresponds to a normalized 8 h-average sound level of approximately 85 dB; a sound exposure of 3,20 $\mathrm{Pa^2h}$ corresponds to a normalized 8 h-average sound level of 90 dB.

Note 6 to entry: Normalized 8 h-average sound level in Formula (2) is identical to "daily noise exposure level" ($L_\mathrm{EX,8h}$) defined in Directive 2003/10/EC, Article 2.

Note 7 to entry: Normalized 8 h-average sound level in Formula (2) is also the same as "noise exposure level normalized to a nominal 8 h working day, $L_\mathrm{EX,8\,h}$" defined in ISO 1999.

3.3
exchange rate
change in sound level corresponding to a doubling or halving of the duration of a steady input sound pressure signal while a constant percentage non-equal-energy sound exposure is maintained

3.4
non-equal-energy sound exposure
time integral of a power function of the time-weighted square of a frequency-weighted sound pressure signal over a stated time interval or event of stated duration

Note 1 to entry: Duration of integration is included implicitly in the time integral and is not always reported explicitly, although it is useful to state the nature of the event. For measurements of non-equal-energy sound exposure over a specified time interval, duration of integration is usually reported and indicated by a suitable subscript to the letter symbol, for example as $E_\mathrm{neeA,1\,h}$.

Note 2 to entry: In symbols and as an example, A-frequency weighted F-time weighted non-equal-energy sound exposure $E_{\mathrm{neeAF},T}$ is represented by Formula (6):

$$E_{\mathrm{neeAF},T} = \int_{t_1}^{t_2}\left(\left(\left(1\big/\tau_\mathrm{F}\right)\int_{-\infty}^{t} p_\mathrm{A}^2\left(\xi\right)\mathrm{e}^{-\left(t-\xi\right)/\tau_\mathrm{F}}\,\mathrm{d}\xi\right)^{10\lg(2)/Q}\right)\mathrm{d}t \tag{6}$$

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where

τ_F is the exponential time constant in seconds for time weighting F;

ξ is a dummy variable of time integration from some time in the past, as indicated by $-\infty$ for the lower limit of the integral, to the time of observation t;

pA(ξ) is the A-weighted instantaneous sound-pressure signal;

Q is the exchange rate in decibels;

T is the integration time beginning at t_1 and ending at t_2.

Note 3 to entry: The unit of sound exposure is pascal-squared seconds (Pa^2s) if sound pressure is in pascals and running time is in seconds.

Note 4 to entry: For applications such as measurement of exposure to noise in the workplace, sound exposure in pascal-squared hours is more convenient than pascal-squared seconds.

Note 5 to entry: If the exchange rate were 3 dB and there were no time weighting, non-equal-energy sound exposure would be equivalent to sound exposure.

3.5
criterion duration
specified duration of time integration

Note 1 to entry: The criterion duration is typically chosen to be the maximum duration of exposure for the application, for example the duration of a working day.

3.6
criterion sound level
specified level of a sound

Note 1 to entry: The criterion sound level is typically chosen to be a level of sound which, if maintained for longer than the criterion duration or if exceeded, could have adverse effects on human health.

3.7
criterion sound exposure
product of the criterion duration and the mean-square sound pressure corresponding to the criterion sound level

Note 1 to entry: The criterion sound exposure is typically chosen to be a legal limit. Such limits vary between jurisdictions and are subject to change.

3.8
criterion non-equal-energy sound exposure
product of the criterion duration and a power function of the mean-square sound pressure corresponding to the criterion sound level

Note 1 to entry: The criterion non-equal-energy sound exposure is typically chosen to be a legal limit. Such limits vary between jurisdictions and are subject to change.

3.9
percentage criterion sound exposure
quotient of sound exposure to criterion sound exposure, multiplied by 100

Note 1 to entry: The unit of percentage criterion sound exposure is per cent.

Note 2 to entry: In symbols and as an example, the percentage criterion sound exposure using frequency weighting A can also be calculated from Formula (7):

$$D_A = \left(100/T_{cr}\right)\int_{t_1}^{t_2} 10^{0,1\left(L_A - L_{cr}\right)}\,dt \qquad (7)$$

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where

T_{cr} is the criterion duration;

T is the integration time beginning at t_1 and ending at t_2;

L_A is the A-frequency weighted sound level equivalent to the A-frequency-weighted input sound pressure;

L_{cr} is the criterion sound level.

3.10
percentage criterion non-equal-energy sound exposure
quotient of non-equal-energy sound exposure to criterion non-equal-energy sound exposure, multiplied by 100

Note 1 to entry: The unit of percentage criterion non-equal-energy sound exposure is per cent.

Note 2 to entry: In symbols and as an example, the percentage criterion non-equal-energy sound exposure using frequency weighting A and time weighting S can also be calculated from Formula (8):

$$D_{ASQ} = \left(100/T_{cr}\right)\int_{t_1}^{t_2} 10^{\left[\left(L_{AS}-L_{cr}\right)\left(\lg(2)/Q\right)\right]}\mathrm{d}t \tag{8}$$

where

Q is the exchange rate in decibels;

T_{cr} is the criterion duration;

T is the integration time beginning at t_1 and ending at t_2;

L_{AS} is the S-time-weighted A-frequency-weighted sound level equivalent to the A-frequency-weighted input sound pressure;

L_{cr} is the criterion sound level.

Note 3 to entry: If the exchange rate were 3 dB and there were no time weighting, percentage criterion non-equal-energy sound exposure would be equivalent to percentage criterion sound exposure.

3.11
time-averaged non-equal-energy sound level
average sound level during the integration time over which a percentage criterion non-equal-energy sound exposure is measured

Note 1 to entry: In symbols and as an example, the time-averaged non-equal-energy sound level using frequency weighting A and time-weighting S can be calculated from Formula (9):

$$L_{neeAS,T} = L_{cr} + \left(Q/\lg(2)\right)\lg\left(D_{ASQ}T_{cr}/100T\right) \tag{9}$$

where

Q is the exchange rate in decibels;

D_{ASQ} is the percentage criterion non-equal-energy sound exposure;

T_{cr} is the criterion duration;

T is the integration time;

L_{cr} is the criterion sound level.

Note 2 to entry: If the sound level is held constant during a test, time-averaged non-equal-energy sound level is that constant level.

Note 3 to entry: If the exchange rate were 3 dB and there were no time weighting, time-averaged non-equal-energy sound level is a close approximation of time-averaged sound level.

Note 4 to entry: Time-averaged non-equal-energy sound level is equivalent to average sound level as described in ANSI S1.25-1991, 4.7.

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4 Performance specifications

4.1 General performance requirements

4.1.1 Components

Generally, a personal sound exposure meter is a combination of a microphone, a signal processor, and a display device. Performance specifications of this document apply to any design of microphone that is appropriate for a personal sound exposure meter.

NOTE 1 The signal processor typically includes the combined functions of an amplifier with a specified and controlled frequency response, a device to form the square of the frequency-weighted, time-varying sound-pressure signal, and a time integrator.

NOTE 2 Computer software is an integral part of some personal sound exposure meters.

All components that are necessary for conformance to the specifications are integral parts of a personal sound exposure meter.

The manufacturer shall provide a means to apply an electrical input signal in place of the signal from the microphone, for the purpose of performing tests on the complete instrument without the microphone.

NOTE 3 A combination of limited acoustical signal tests with electrical input signal tests can be more practicable than acoustical signal tests of the entire personal sound exposure meter.

The manufacturer may provide an accessible input test point or recommend and provide a dummy microphone or equivalent input adaptor (electrical or non-electrical) for performing electrical tests on the instrument. The point of access for electrical input signals should be before any signal processing such that electrical input signals for testing can be applied to as much of the personal sound exposure meter as possible.

EXAMPLE 1 Some personal sound exposure meters allow the microphone to be removed to allow electrical signals to be applied to the remainder of the personal sound exposure meter.

An optional extension device or cable may be provided by the manufacturer of the personal sound exposure meter for installation between the components. If such device or cable is provided, details shall be given in the instruction manual for any corrections to be applied to the results of measurements made in this manner (see 4.27.4).

EXAMPLE 2 Some personal sound exposure meters allow for an extension cable for installation between the microphone or a microphone preamplifier and the other components.

Accessories including windscreens or rain protection devices may be specified for installation around the microphone (see 4.11.2).

It is recommended that the manufacturer provide an accessible output test point (see 4.22).

NOTE 4 An output test point such as an AC analogue signal port or digital output port enables easier determination of conformance to specifications.

4.1.2 Display device

The display device shall provide a physical and visible display, or storage, of measurement results. Any stored measurement result shall be available for display by means of a manufacturer-specified device. An analogue or digital output connection alone is not a display device.

EXAMPLE Some personal sound exposure meters use a computer with associated software to provide the display.

The display device may be built into, or be separate from, the wearable part of the personal sound exposure meter.

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The acoustical quantity that is being measured shall be clearly indicated on the display or marked near the display. The indication may be by means of a suitable letter symbol or an abbreviation. Examples of appropriate letter symbols are given with the definitions, formulas, and tables in this document.

A personal sound exposure meter may have more than one display device. Each alternative device for displaying the signal level, stated in the instruction manual as conforming to the specifications of this document, is an integral part of the personal sound exposure meter. Each such alternative device shall be included as part of the components required for demonstration of conformance to the performance specifications in Clause 4.

The display range for sound levels shall be at least 70 dB. The resolution of displayed sound levels shall not exceed 0,1 dB.

If sound exposure is provided, the display shall indicate sound exposure, either as a direct indication or as a fraction or a percentage of a manufacturer-specified sound exposure. If sound exposure is not indicated directly in pascal-squared hours, the manufacturer shall provide a suitable means to convert the indication to sound exposure in pascal-squared hours.

If percentage criterion sound exposure is provided, the applicable criterion sound level and criterion duration shall be clearly indicated.

If percentage criterion non-equal-energy sound exposure is provided, the applicable criterion sound level, criterion duration, and exchange rate shall be clearly indicated.

The resolution of displayed sound exposures should not exceed 0,01 Pa^2h. If the displayed sound exposure is less than 0,1 Pa^2h, the sound exposure may be displayed in units of pascal-squared seconds.

The resolution of displayed percentage criterion sound exposures and percentage criterion non-equal-energy sound exposures shall not exceed 1 %.

For digital display devices updated at periodic intervals, the indication at each display update shall be the value of the user-selected quantity at the time of the display update. Other quantities may be indicated at the time of the display update.

4.1.3 Configuration

The manufacturer shall specify the configuration of the complete personal sound exposure meter and its normal mode of operation for conformance to the requirements of this document.

If appropriate, the configuration of the complete personal sound exposure meter may include a windscreen and other devices that are installed around the microphone as integral components for the normal mode of operation.

NOTE In use, the characteristics of a personal sound exposure meter are strongly affected by the presence of the wearer of the instrument and the mounting position of the instrument or its microphone.

4.1.4 Design features

4.1.4.1 General

A personal sound exposure meter may provide any or all of the design features for which performance specifications are given in this document. A personal sound exposure meter shall conform to the applicable performance specifications for those design features that are provided.

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Performance specifications in this document apply, as applicable, to any time or frequency weightings operated in parallel and to each independent channel of a multi-channel personal sound exposure meter. A multi-channel personal sound exposure meter may have two or more microphone inputs.

Personal sound exposure meters shall conform to the specifications of this document under the reference environmental conditions given in 4.1.7.

Specifications of this document apply after an initial time interval following switching on the power to the personal sound exposure meter. The initial time interval shall not exceed 2 min. The personal sound exposure meter shall be allowed to reach equilibrium with the prevailing ambient environment before switching on the power.

A personal sound exposure meter may provide design features (for example, a facility to record a time history of time-averaged sound levels) for which performance specifications are not given in this document. Such design features are not tested at pattern evaluation or periodic verification.

4.1.4.2 Mandatory characteristics

A personal sound exposure meter shall provide a means to indicate

– the A-frequency-weighted time-averaged sound level,
– the C-frequency-weighted time-averaged sound level, and
– the C-frequency-weighted peak sound level.

A hold feature shall be provided for measurements of peak sound level.

4.1.4.3 Optional characteristics

4.1.4.3.1 Sound exposure

A personal sound exposure meter may provide a means to indicate sound exposure.

If so, the frequency weighting shall be A or C as specified by IEC 61672-1.

Sound exposures that have been accumulated during a measurement period shall be retained in the memory until the instrument is reset and shall not be deleted by triggering of the latching overload indicator.

4.1.4.3.2 Sound exposure level

A personal sound exposure meter may provide a means to indicate sound exposure level.

If so, the frequency weighting shall be A or C as specified by IEC 61672-1.

4.1.4.3.3 Percentage criterion sound exposure

A personal sound exposure meter may provide a means to indicate percentage criterion sound exposure.

If so, the frequency weighting shall be A or C as specified by IEC 61672-1.

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4.1.4.3.4 Percentage criterion non-equal-energy sound exposure

A personal sound exposure meter may provide a means to indicate percentage criterion non-equal-energy sound exposure.

If so,

– the time weighting shall be F or S as specified by IEC 61672-1,
– the exchange rate shall be 3 dB, 4 dB, or 5 dB,
– frequency weighting A as specified by IEC 61672-1 shall be provided, and
– frequency weighting C as specified by IEC 61672-1 may also be provided.

4.1.4.3.5 Time-averaged non-equal-energy sound level

If a personal sound exposure meter provides a means to indicate percentage criterion non-equal-energy sound exposure, it shall also provide a means to indicate time-averaged non-equal-energy sound level.

For time-averaged non-equal-energy sound level,

– the time weighting shall be F or S as specified by IEC 61672-1,
– the exchange rate shall be 3 dB, 4 dB, or 5 dB,
– frequency weighting A as specified by IEC 61672-1 shall be provided, and
– frequency weighting C as specified by IEC 61672-1 may also be provided.

4.1.4.3.6 Time-weighted sound level

A personal sound exposure meter may provide a means to indicate frequency-weighted time-weighted sound level.

If so,

– frequency weighting A and time weighting F as specified by IEC 61672-1 shall be provided,
– frequency weighting C and time weighting S as specified by IEC 61672-1 may also be provided, and
– a hold feature shall be provided for measurements of maximum time-weighted sound level.

4.1.5 Adjustment at the calibration check frequency

A personal sound exposure meter shall provide a means for the user to check and adjust its overall sensitivity at the calibration check frequency using a sound calibrator in order to optimise its electroacoustical performance over the complete frequency range.

At least one model of sound calibrator shall be stated in the instruction manual for checking or adjusting the overall sensitivity. Any adaptor used with the sound calibrator shall be clearly identified and referred to in the statement of the adjustment data. The sound calibrator shall conform to either the class 1 or the class 2 specifications of IEC 60942. Conformance shall be demonstrated for sound pressure level, frequency, and distortion using the methods given in IEC 60942. Evidence that the sound calibrator conforms to the applicable requirements of IEC 60942 may be provided by documentation from a periodic test performed in accordance with the procedures given in IEC 60942.

NOTE 1 Class LS (laboratory standard) sound calibrators are not suitable for general field applications with personal sound exposure meters because their performance characteristics are specified in IEC 60942 only for a limited range of environmental conditions.

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The reference sound pressure level for testing of electroacoustical performance should preferably be 94 dB or 114 dB.

NOTE 2 Sound pressure levels of 94 dB and 114 dB correspond closely to the levels of time-mean-square sound pressures of 1 Pa^2 and 100 Pa^2 respectively or root-mean-square sound pressures of 1 Pa and 10 Pa respectively.

For the reference sound pressure level and for a calibration check frequency in the range from 160 Hz to 1 250 Hz, a procedure and data shall be provided in the instruction manual so that the adjustment applied yields the appropriate indication of time-averaged sound level in response to the application of the sound calibrator.

The diffuse-field adjustment data for a personal sound exposure meter shall be determined using a diffuse-field method. This diffuse-field method should be based on the free-field methods for determination of adjustment data for a sound level meter given in IEC 62585:2012, Clause 6, Clause 8, and Annex A.

The adjustment data shall apply for environmental conditions at least within the ranges of 80 kPa to 105 kPa for static pressure, 20 °C to 26 °C for air temperature, and 25 % to 70 % for relative humidity. The adjustment data shall apply for the configuration of the personal sound exposure meter stated in the instruction manual.

The adjustment data have no associated uncertainties of measurement.

The difference between the adjustment data measured according to Clause 5 and the adjustment data from the instruction manual shall not exceed ±0,3 dB.

4.1.6 Determination of conformance

The personal sound exposure meter shall conform to all applicable performance specifications that are given in this document for those design features which are provided. Where applicable, acceptance limits are provided for allowable values of measured deviations from design goals. Annex B describes the relationship between tolerance interval, corresponding acceptance interval, and the maximum-permitted uncertainty of measurement.

Conformance to a performance specification is demonstrated when the following criteria are both satisfied:

a) measured deviations from design goals do not exceed the applicable acceptance limit;
b) the corresponding uncertainty of measurement does not exceed the corresponding maximum-permitted uncertainty of measurement given in Annex C for a coverage probability of 95 %.

Annex D gives examples of evaluation of conformance to the specifications of this document.

4.1.7 Reference environmental conditions

Electroacoustic performance specifications apply under the following reference environmental conditions:

– air temperature: 23 °C;
– static pressure: 101,325 kPa;
– relative humidity: 50 %.

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4.1.8 Classification for radio-frequency tests

For specifying the maximum-permitted emission of, and immunity to the effects of exposure to, radio-frequency fields, personal sound exposure meters are classified into two groups as follows:

– group X personal sound exposure meters: self-contained instruments that include sound level measurement facilities according to this document and which specify internal battery power for the normal mode of operation, requiring no external connections to other apparatus to measure sound levels;
– group Z personal sound exposure meters: instruments that include sound level measurement facilities according to this document and which require two or more items of equipment which are essential components of the personal sound exposure meter, to be connected together by any means for the normal mode of operation. The separate items may be operated from batteries or from a public supply of electric power.

NOTE The group X and Z designations are equivalent to those in IEC 61672-1 and IEC 61672-2.

4.2 Marking

A personal sound exposure meter that is designed to conform to all the applicable specifications of this document shall be marked to show

– the IEC number of this document (IEC 61252) and the applicable year of publication,
– the supplier responsible for the technical specifications applicable to the complete personal sound exposure meter, and
– the model designation and a serial number.

NOTE Sometimes the manufacturer of the personal sound exposure meter is the supplier responsible for the technical specifications.

If the personal sound exposure meter consists of several separate units, each principal unit or component shall be marked as described in 4.2, as practicable.

If the personal sound exposure meter is powered by batteries that are intended to be user-replaceable, the acceptable battery types should be stated on the instrument.

A personal sound exposure meter may also be marked to show the quantity(ies) that it displays and associated information.

EXAMPLE If the personal sound exposure meter displays an indication of percentage criterion sound exposure or percentage criterion non-equal-energy sound exposure, appropriate associated information is the sound exposure or non-equal-energy sound exposure in pascal-squared hours that corresponds to 100 %.

4.3 Static pressure

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 6.1 and 6.2.

4.4 Temperature

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 6.1 and 6.3, except that the acceptance limits shall be ±0,5 dB.

NOTE 1 The acceptance limits are equivalent to those given in IEC 61252:1993.

NOTE 2 IEC 61672-1:2013, 6.3, permits some components to be designated as intended to operate only in an environmentally controlled enclosure and therefore permits a more restricted range of air temperature for operation of those components.

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4.5 Relative humidity

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 6.1 and 6.4, except that the acceptance limits shall be ±0,5 dB.

NOTE The acceptance limits are equivalent to those given in IEC 61252:1993.

4.6 Electrostatic discharge

A personal sound exposure meter shall conform to the requirements for a sound level meter of IEC 61672-1:2013, 6.1 and 6.5.

4.7 AC-power-frequency and radio-frequency fields

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 6.1 and 6.6.

4.8 Radio-frequency emissions and disturbances to a public power supply

A personal sound exposure meter shall conform to the applicable requirements of IEC 61672-1:2013, 5.21.

4.9 Directional response

At any frequency in the range of a personal sound exposure meter, the directional response design goal is equal response to sounds from all directions of sound incidence. Table 1 specifies acceptance limits for deviations from the design goal as limits on the maximum absolute value of the difference between displayed sound levels at any two sound-incidence angles within certain angular regions around the reference direction.

NOTE 1 IEC 61252:1993 did not provide specifications for directional response of personal sound exposure meters.

The directional-response requirements of Table 1 apply for the configuration of a personal sound exposure meter as stated in the instruction manual for the normal mode of operation or for those components of a personal sound exposure meter that are intended to be located in a sound field, but not mounted on a person and without an observer disturbing the sound field.

NOTE 2 In use, the directional characteristics of a personal sound exposure meter are strongly affected by the presence of the wearer of the instrument and the mounting position of the instrument or its microphone. The specifications in 4.9 are intended to enable consistent testing for all designs of personal sound exposure meter.

The specifications in Table 1 apply for sinusoidal progressive sound waves at any sound-incidence angle within the specified ranges, including the reference direction and in any plane containing the principal axis of the microphone.

For any frequency within the specified ranges, the requirements of Table 1 apply for any orientation of the personal sound exposure meter, or applicable components, around the reference direction. The requirements of Table 1 apply for indications of any frequency-weighted sound levels.

For any pair of sound levels displayed within each range of sound-incidence angles in Table 1, and at any frequency in a specified range, the measured absolute values of the differences between the displayed sound levels shall not exceed the applicable limits given in Table 1.

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Table 1 – Acceptance limits for deviations of directional response from the design goal

Frequency kHz	Maximum absolute value of the difference between displayed sound levels at any two sound-incidence angles within ±θ° from the reference direction dB		
	θ = 30°	θ = 90°	θ = 150°
0,25 to 2	2,0	6,0	7,0
> 2 to 4	4,0	7,0	12,0
> 4 to 8	6,0	17,0	17,0

It is recommended that detailed tables of directional response (see 4.27.2) be provided in the instruction manual of the personal sound exposure meter.

NOTE 3 The provision of tables of directional response in the instruction manual enables simpler measurements of directional response during pattern evaluation.

4.10 Frequency weightings

For all frequency weightings, the design goals and acceptance limits for the class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.5, apply for the relative frequency-weighted diffuse-field response.

NOTE For the purposes of this document, the diffuse-field response is equivalent to the random-incidence response and so the requirements of IEC 61672-1 are applicable.

The requirements for frequency weighting apply for the configuration of a personal sound exposure meter as stated in the instruction manual for the normal mode of operation or for those components of a personal sound exposure meter that are intended to be located in a sound field but not mounted on a person and without an observer disturbing the sound field.

4.11 Corrections to indicated levels

4.11.1 General

Corrections from the instruction manual for the influence of various effects may be used in measurements of sound level and sound exposure by a user and in tests of the performance of the personal sound exposure meter.

Correction data and the associated uncertainties of measurement for a coverage probability of 95 %, together with the associated coverage factor, as appropriate, shall be determined using a diffuse-field method. This diffuse-field method should be based on the free-field methods for determination of correction data given in IEC 62585 for a class 2 sound level meter. The methods of IEC 61183 shall be used to determine the diffuse-field response of a personal sound exposure meter.

NOTE For the purposes of this document, diffuse-field sound level is equivalent to random-incidence sound level and so both the random-incidence and diffuse-field methods of IEC 61183 are suitable for determining the diffuse-field response of a personal sound exposure meter.

Correction data and the associated uncertainties of measurement shall be given separately in tabular form in the instruction manual. The uncertainties given in the instruction manual shall represent real and realistic (non-zero) uncertainties, even if a correction is zero. The uncertainties shall not exceed the applicable corresponding maximum-permitted uncertainties for free-field corrections given in IEC 62585:2012, Clauses 10, 12, 13, and 14, as required, for a class 2 sound level meter.

Corrected results shall be obtained by adding the appropriate correction data to indicated levels. The methods and criteria that shall be used to validate the correction data for the purpose of pattern evaluation are given in 5.11.

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4.11.2 Windscreens

The instruction manual shall provide corrections for the average effects of a windscreen on directional response and on the relative frequency-weighted diffuse-field response of the personal sound exposure meter.

Windscreen-correction data are required if the instruction manual states that the personal sound exposure meter conforms to the specifications of this document both in a configuration that includes a windscreen and in a configuration that does not include a windscreen.

When a windscreen and its associated accessories are not rotationally symmetric about the principal axis of the microphone, diffuse-field correction data for the effect of the windscreen and accessories on directional response and frequency response shall be provided for various sound-incidence angles in appropriate planes through the principal axis of the microphone.

The correction data for directional response shall be provided in tabular form at one-third-octave intervals for nominal frequencies from 63 Hz to at least 8 kHz.

As required, corrections for the average effects of a stated type of windscreen on the relative frequency-weighted diffuse-field response of the personal sound exposure meter in the reference direction shall be provided in tabular form at one-third-octave intervals for nominal frequencies from 1 kHz to 8 kHz.

IEC 61672-1:2013, Annex D, gives frequencies at one-third-octave intervals.

The difference between a windscreen correction measured during a pattern-evaluation test of frequency weighting (see 5.10) and the corresponding windscreen correction given in the instruction manual shall not exceed ±0,5 dB for nominal frequencies from 63 Hz to 2 kHz and shall not exceed ±0,8 dB for nominal frequencies greater than 2 kHz to 8 kHz.

4.11.3 Corrections for use during periodic testing

If the instruction manual recommends a multi-frequency sound calibrator, a comparison coupler, or an electrostatic actuator for acoustical signal tests of a frequency weighting of a personal sound exposure meter during periodic verification (see 6.3.1), the instruction manual shall provide correction data to obtain frequency-weighted sound levels equivalent to those that would be displayed under reference environmental conditions in response to a diffuse sound field.

Electrostatic actuators shall conform to the requirements of IEC 61094-6.

The correction data for use during periodic testing shall be provided at least for frequencies of 31,5 Hz, 1 kHz, 4 kHz, and 8 kHz and shall apply for stated configurations of a personal sound exposure meter, and a model of sound calibrator, comparison coupler, or electrostatic actuator. The correction data shall be provided for all models of microphones or microphone-windscreen configurations for which the personal sound exposure meter is stated in the instruction manual to conform to the specifications of this document. The uncertainties of the correction data shall be provided for at least the above-mentioned frequencies and configurations.

Deviations of the relative equivalent frequency response, determined from application of the sound calibrator, comparison coupler, or electrostatic actuator, from the relative frequency-weighted diffuse-field response, measured according to 5.11, shall not exceed the values of uncertainty given in the instruction manual for the correction data.

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4.12 Linearity of response to steady signals

The extent of the linear operating range shall be at least from 67 dB to 137 dB for nominal A-weighted sound levels at 1 kHz.

NOTE For a sinusoidal signal, a sound level of 137 dB corresponds to a peak sound level of 140 dB.

For the entire level range, a personal sound exposure meter shall conform to the applicable class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.6. If the stated range of sound level exceeds the minimum requirements specified in 4.12, all specifications and associated tolerances shall apply to the ranges stated.

4.13 Under-range indication

A personal sound exposure meter shall conform to the applicable requirements for a sound level meter of IEC 61672-1:2013, 5.12.

NOTE In use, the under-range condition is rarely displayed during measurements of sound exposure level because sound exposure is integrated over time.

4.14 Time-weightings F and S

If the personal sound exposure meter provides time-weighted sound level (see 4.1.4.3.6), the personal sound exposure meter shall conform to the applicable requirements for a sound level meter of IEC 61672-1:2013, 5.8.

4.15 Response to short duration signals for indications of time-averaged sound levels, sound exposure levels, and time-weighted sound levels

A personal sound exposure meter shall conform to the applicable class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.9 and 5.10, for indications of time-averaged sound levels and, if provided, sound exposure levels and time-weighted sound levels.

4.16 Response to short duration signals for indications of time-averaged non-equal-energy sound levels

For a personal sound exposure meter that provides a means to indicate percentage criterion non-equal energy sound exposure (see 4.1.4.3.4), specifications for response to short-duration signals apply to the indication of time-averaged non-equal-energy sound level (see 4.1.4.3.5) in response to repeated 4 kHz tonebursts.

The specification for the response applies for any sequence of 4 kHz tonebursts of equal amplitude and equal duration. The reference toneburst response to the sequence of tonebursts is given in Table 2 for the applicable combination of time weighting and exchange rate Q.

Measured deviations of time-averaged non-equal-energy sound levels from the corresponding reference time-averaged non-equal-energy sound levels shall not exceed the applicable acceptance limits of Table 2.

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Table 2 – Reference response to a sequence of repeated 4 kHz tonebursts for time-averaged non-equal-energy sound levels and acceptance limits

Toneburst duration ms	Reference response to a sequence of repeated 4 kHz tonebursts relative to the steady sound level dB			Acceptance limits dB
	$L_{neeAF,T} - L_{AF}$; $Q = 3$	$L_{neeAF,T} - L_{AF}$; $Q = 4$	$L_{neeAF,T} - L_{AF}$; $Q = 5$	
1 000	−10,0	−12,9	−15,8	±1,0
500	−13,0	−16,6	−20,1	±1,0
200	−16,9	−21,2	−25,1	±1,0
100	−20,0	−24,4	−28,5	±1,0
50	−23,0	−27,5	−31,7	+1,0; −1,5
20	−26,9	−31,6	−35,8	+1,0; −2,0
10	−30,0	−34,6	−38,9	+1,0; −2,0
5	−33,0	−37,6	−41,9	+1,0; −2,5
2	−36,9	−41,6	−45,9	+1,0; −2,5
1	−40,0	−44,6	−49,0	+1,0; −3,0
0,5	−43,0	−47,6	−52,0	+1,0; −4,0
0,25	−46,0	−50,7	−55,1	+1,5; −5,0
	$L_{neeAS,T} - L_{AS}$; $Q = 3$	$L_{neeAS,T} - L_{AS}$; $Q = 4$	$L_{neeAS,T} - L_{AS}$; $Q = 5$	
1 000	−10,0	−11,4	−12,5	±1,0
500	−13,0	−14,5	−15,7	±1,0
200	−17,0	−18,6	−19,9	±1,0
100	−20,0	−21,6	−22,9	±1,0
50	−23,0	−24,7	−26,0	+1,0; −1,5
20	−27,0	−28,6	−30,0	+1,0; −2,0
10	−30,0	−31,7	−33,0	+1,0; −3,0
5	−33,0	−34,7	−36,0	+1,0; −4,0
2	−37,0	−38,7	−40,1	+1,5; −5,0
NOTE The reference response values were determined by numerical simulation of ideal integration and averaging processes.				

The specification for the response applies for toneburst durations in the range given in Table 2 and for a steady input signal level equivalent to an indication of 3 dB less than the specified upper boundary of the linear operating range at 4 kHz.

4.17 Overload indication

A personal sound exposure meter shall conform to the applicable class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.11, for indications of time-averaged sound levels and peak sound levels and, if provided, sound exposure levels and time-weighted sound levels.

To avoid spurious overload conditions, the overload indicator may include an exponential time constant that does not exceed 1 ms.

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4.18 C-weighted peak sound level

4.18.1 Deviation from reference differences

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.13.

4.18.2 Repeatability

In addition, for a series of ten measurements of C-weighted peak sound level performed at random intervals during a period of 2 min in response to one-cycle electrical signals with a nominal frequency of 8 kHz that start and stop on zero crossings,

– all ten indicated C-weighted peak sound levels shall lie within a range of 2,0 dB, and
– the standard deviation of the ten indicated C-weighted peak sound levels shall not exceed 0,6 dB.

NOTE These requirements are intended to ensure that a personal sound exposure meter samples the input signal at an appropriate rate for consistent indication of C-weighted peak sound level.

4.19 Stability during continuous operation

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.14, for indications of A-frequency-weighted time-averaged sound levels.

4.20 High-level stability

A personal sound exposure meter shall conform to the applicable class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.15, for indications of A-frequency-weighted time-averaged sound levels.

4.21 Reset facility

A personal sound exposure meter shall provide a facility to clear the data storage device and to re-initiate a measurement.

Use of a reset facility shall not cause spurious indications on a display device or errors in data that are stored.

4.22 Analogue or digital output

A personal sound exposure meter shall conform to the applicable class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.19.

4.23 Timing facilities

A personal sound exposure meter shall provide a means for indication of averaging times. The design goal for the relative difference between the indicated internal averaging time and the actual averaging time is 0 % with acceptance limits of ±1 %.

The capability to preset an averaging time interval or an integration time interval, if applicable, may be provided. Recommended preset time intervals are 10 s, 1 min, 5 min, 10 min, 30 min, 1 h, 8 h, and a duration of 24 h. The maximum averaging time interval or integration time interval should not exceed 24 h.

Time of day may also be displayed.

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4.24 Crosstalk

A multi-channel personal sound exposure meter shall conform to the applicable requirements of IEC 61672-1:2013, 5.22.

4.25 Power supply

A personal sound exposure meter shall conform to the class 2 requirements for a sound level meter of IEC 61672-1:2013, 5.23.

A power supply of the type recommended by the manufacturer shall be sufficient for operation of the personal sound exposure meter in conformance with the applicable performance specifications for at least 8 h under reference environmental conditions.

A personal sound exposure meter shall not continue to record data if the power supply voltage falls below the minimum specified in the instruction manual.

4.26 Pause and resumption of time averaging and time integration

A personal sound exposure meter may provide a means for pausing and resuming time-averaging and time-integrating measurements and the corresponding elapsed time.

If so, the numerical value of all time-averaged or time-integrated quantities for which measurement is in progress shall be retained in memory without modification while the measurements are paused. Pause or resumption of measurements shall not cause spurious indications on a display device or errors in any stored data.

4.27 Instruction manual

4.27.1 General

An instruction manual shall be supplied with each personal sound exposure meter that conforms to the specifications of this document.

If the personal sound exposure meter consists of several separate components, an instruction manual shall be available for the combination that forms the complete personal sound exposure meter. The instruction manual shall describe all necessary components and their mutual influence.

4.27.2 Specifications

The instruction manual shall state at least the following information:

a) the configuration of the complete personal sound exposure meter and its normal mode of operation for conformance to the requirements of this document (see 4.1.3);
b) the classification group X or Z for immunity to the effects of exposure to radio-frequency fields (see 4.1.8);
c) the models of microphones with which the complete personal sound exposure meter conforms to the requirements of this document;
d) how the microphone and preamplifier are to be mounted, if applicable, to conform to the specifications for directional response and frequency weightings. If an extension device or cable is required to conform to the specifications, the personal sound exposure meter shall be stated as conforming to the applicable specifications for directional response and frequency weighting only when the specified device(s) or cable(s) are installed;

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e) as appropriate for each specified model of microphone, the design goal and applicable acceptance limits for either the electrical characteristics of the device, or the means, used to insert electrical input signals for electrical tests of the instrument (see 4.1.1); electrical characteristics include the resistive and reactive components of the electrical impedance at the output of the device; the design goal for the impedance shall be specified for a frequency of 1 kHz;

f) a means to identify the version of all software that is integral to operation of the personal sound exposure meter;

g) a description of all display devices, including the modes of operation and applicable display-update rates for digital displays; if more than one display device is provided, a statement as to which of these devices conform to the specifications of this document and which are for other purposes (see 4.1.2);

h) description of the acoustical quantities that the personal sound exposure meter is capable of measuring on each display device, for example time-averaged sound level, sound exposure level, or time-weighted sound level, separately or in combinations, along with explanations of all abbreviations, letter symbols, and icons that are displayed (see 4.1.4);

i) if a personal sound exposure meter is capable of displaying sound exposure level, the value of the reference sound exposure;

j) if percentage criterion sound exposure (see 4.1.4.3.3) is provided, the associated frequency weighting, the criterion sound level or criterion sound exposure, and criterion duration;

k) if percentage criterion non-equal-energy sound exposure (see 4.1.4.3.4) is provided, the associated time weighting, exchange rate, frequency weighting, the criterion sound level or criterion sound exposure, and criterion duration;

l) description of all frequency weightings that conform to the specifications of this document (see 4.10);

m) identification of the upper and lower frequencies for which the personal sound exposure meter is designed to operate if these are outside the range required by 4.10;

n) description of the time weightings that are provided (see 4.14);

o) identification of the level range by the nominal A-weighted sound levels at the lower and upper boundaries of the linear operating range at 1 kHz (see 4.12);

p) the range of C-weighted peak sound levels that may be measured;

q) the design-goal characteristics and the acceptance limits that should be maintained for quantities that the personal sound exposure meter is capable of indicating but for which no performance specifications are provided in this document, including any optional frequency weightings;

r) the reference sound pressure level, reference level range, and reference orientation;

s) the reference direction and the location of the microphone reference point for each model of microphone intended for use with the personal sound exposure meter;

t) for signal levels within the range of a display device, the minimum and maximum averaging and integration times for measurement of time-averaged sound levels and sound exposure levels, if provided, respectively;

u) if a digital indicator is provided, the display update rate and the conditions to be achieved after initiating a measurement when the first valid indication is displayed;

v) if an analogue or digital output is provided, a description of the characteristics of the output; for analogue outputs, the characteristics shall include the frequency weighting, the range of root-mean-square voltages for sinusoidal output signals, the internal electrical impedance at the output, and the recommended range of load impedances;

w) if the personal sound exposure meter is designed to accept the connection of interface or interconnection cables, the lengths and types of typical cables (for example, shielded or unshielded) and the characteristics of all devices to which the cables are expected to be attached;

x) the maximum and minimum power supply voltages at which the personal sound exposure meter conforms to the specifications of this document;

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y) if internal batteries are used to power the personal sound exposure meter, recommendations for acceptable battery types and the nominal duration of continuous operating, under reference environmental conditions, to be expected for the specified normal mode of operation when full capacity batteries are installed;

z) for personal sound exposure meters powered by internal batteries and designed to be able to indicate sound levels over a duration that exceeds the nominal battery life, a description of the recommended means for operating the personal sound exposure meter from an external power supply;

aa) the method to confirm that the power supply is sufficient to operate the personal sound exposure meter in conformance to the specifications of this document (see 4.25);

bb) for personal sound exposure meters that are intended to operate from a public supply of AC electrical power, a statement of the nominal root-mean-square voltage and frequency of the power supply and the acceptance limits around the nominal values;

cc) the typical time interval needed for the personal sound exposure meter to stabilize after changes in environmental conditions;

dd) identification of the components of the personal sound exposure meter intended to be operated only in an environmentally controlled enclosure (see 4.4);

ee) the effects of electrostatic discharges on the operation of the personal sound exposure meter; a statement of the temporary degradation or loss, if any, in the performance or function of the personal sound exposure meter resulting from exposure to electrostatic discharges; for personal sound exposure meters that require internal access for maintenance by a user, a statement, if needed, of precautions against damage by electrostatic discharges;

ff) a statement that the personal sound exposure meter conforms to the basic specification of this document for the required immunity to AC power-frequency and radio-frequency fields; alternatively, if applicable, a statement of the F-time-weighted or time-averaged sound levels less than 74 dB, on all applicable level ranges, for which the personal sound exposure meter conforms to the specifications of this document upon exposure to AC power-frequency and radio-frequency fields; the information shall specify the frequency of the AC power field (see 4.7).

The instruction manual should also state the following information:

gg) tabulated values of directional response for the personal sound exposure meter in the configuration for the normal mode of operation as functions of sound incidence angle and frequency, of the response relative to the corresponding response in the reference direction (see 4.9); the frequencies shall be from 250 Hz to 2 kHz at nominal one-third-octave frequencies and from greater than 2 kHz to 8 kHz at nominal one-sixth-octave intervals (see IEC 61672-1:2013, Annex D, for the nominal frequencies); at each frequency, angular intervals for the tabulated values of directional response shall not exceed 10°;

hh) if the personal sound exposure meter can display the time of day, the nominal drift in the displayed time over a period of 24 h;

ii) if the personal sound exposure meter conforms to the specifications of this document at unmodulated, root-mean-square electric field strengths greater than the specified field strengths, the applicable field strengths.

4.27.3 Information for operation

The instruction manual shall describe at least the following information:

a) appropriate procedures for use of the personal sound exposure meter;

b) the means by which a user can identify the version of any software that is required to operate the personal sound exposure meter;

c) the placement of the personal sound exposure meter on the person;

d) the greatest sound pressure level at the microphone that can be applied without causing damage to the personal sound exposure meter;

e) the operation of the hold facility and the means for clearing a display that is held;

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f) the characteristics and operation of each independent channel of a multi-channel personal sound exposure meter;

g) procedure for selecting the acoustical quantity to be indicated, including choice of frequency weighting, time weighting, and exchange rate, as applicable;

h) if applicable, the procedure to preset an averaging- or integration-time interval and to set the time of day;

i) model(s) of sound calibrator for checking or adjusting the overall sensitivity, including any adaptor to be used with the sound calibrator (see 4.1.5);

j) corrections and the associated uncertainties for the average effects of a windscreen on directional response and on the relative frequency-weighted diffuse-field response for the personal sound exposure meter (see 4.11.2);

k) the operation of the reset facility for measurements of time-averaged sound level, sound exposure level, maximum time-weighted sound level, and peak sound level; a statement as to whether operation of the reset facility clears an overload indication; the nominal delay time between operation of the reset facility and re-initiation of a measurement;

l) the operation and interpretation of overload and under-range indications and the means for clearing the indications;

m) if user-selectable thresholds are provided, the performance and operation of the thresholds;

n) the method to transfer or download digital data to an external data storage or display device and identification of the software and hardware to accomplish those tasks;

o) means to minimize the influence of mechanical vibration on indicated sound levels and sound exposures, if provided;

p) guidance and procedures for use of the personal sound exposure meter at locations or under conditions where the static pressure is less than 85 kPa.

4.27.4 Information for use with auxiliary devices

The instruction manual shall provide at least the following information:

a) if an optional extension device or cable is provided for installation between the components of a personal sound exposure meter, any corrections to be applied to the results of measurements made in this manner;

b) the typical effect on electroacoustical performance when optional accessories supplied by the manufacturer of the personal sound exposure meter are used (see 4.1.1); the data shall apply to all relevant characteristics of the personal sound exposure meter that are affected by installation of the accessories; for any recommended type of windscreen, data shall be provided for the typical effect of the accessory, in the absence of wind, on microphone sensitivity, directional response, and frequency weighting (see 4.11.2);

c) a statement as to whether the personal sound exposure meter conforms to the specifications of this document when an optional accessory is installed;

d) if internal or external bandpass filters are provided for spectral analysis of a sound-pressure signal, a description of how the personal sound exposure meter is to be used to measure filtered sound pressure levels;

e) details of the connection of manufacturer-provided auxiliary devices to a personal sound exposure meter and of the effects, if any, of such devices on the electroacoustical characteristics of the personal sound exposure meter.

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4.27.5 Information for testing

The instruction manual shall provide at least the following information:

a) directivity indexes for determining relative frequency-weighted random-incidence response (see 4.10);

b) for the A-weighted sound levels displayed (1) in response to the sound pressure produced by a calibrated multi-frequency sound calibrator, or (2) as the indications obtained by use of a comparison coupler, or (3) in response to simulation of sound pressure by an electrostatic actuator, correction data to obtain A-weighted sound levels equivalent to the diffuse-field response; correction data and the associated expanded uncertainties shall be supplied at least for the frequencies required for periodic testing (see 4.11.3); the model of sound calibrator, comparison coupler, or electrostatic actuator for which the correction data are valid shall be identified;

c) tables of the nominal A-weighted sound levels and the nominal C-weighted sound levels at the upper and lower boundaries of the linear operating range at least at frequencies of 31,5 Hz, 1 kHz, 4 kHz, and 8 kHz (see 4.12); sound levels should preferably be tabulated for all frequency weightings available in the personal sound exposure meter;

d) for each frequency for which A-weighted sound levels are specified at the lower and upper boundaries of the linear operating ranges, the starting point at which to begin tests of level linearity deviation; at 1 kHz, the starting point shall be the reference sound pressure level;

e) for each model of microphone for which the personal sound exposure meter is stated to conform to the specifications, the greatest sound pressure level at the microphone and the greatest peak-to-peak voltage that can be applied at the electrical input, which the personal sound exposure meter is designed to accommodate;

f) if applicable, the unmodulated, root-mean-square electric field strength greater than the specified field strengths for which the personal sound exposure meter conforms to the specifications of this document;

g) the mode(s) of operation of the personal sound exposure meter, and any connecting devices, which produce the greatest radio-frequency emissions; a description of the configurations of the personal sound exposure meter that produce the same, or lower, radio-frequency emission levels;

h) the operating mode(s) of the personal sound exposure meter, and of any connection devices, which have the least immunity to the effects of exposure to AC power-frequency and radio-frequency fields, and the corresponding reference orientation of the personal sound exposure meter relative to the principal direction of an electromagnetic field;

i) for a multi-channel personal sound exposure meter, details of any termination devices that are installed in place of the microphones on the other inputs.

5 Pattern-evaluation tests

5.1 General requirements

5.1.1 Submission for testing

At least three specimens of the same pattern of personal sound exposure meter shall be submitted for pattern-evaluation testing. As a minimum, the laboratory shall select two of the specimens for testing. At least one of the two specimens shall then be tested fully according to the procedures of Clause 5. The laboratory shall decide whether the full tests shall also be performed on the second specimen or whether additional limited testing is adequate to approve the pattern.

An instruction manual and all items or accessories that are identified in the instruction manual as integral components for the normal mode of operation shall be submitted along with the three personal sound exposure meters. Examples of additional items or accessories include a microphone extension device or cable and peripheral equipment.

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If the manufacturer of the personal sound exposure meter supplies devices that are to be connected to the personal sound exposure meter by cables, then the devices and cables shall be submitted with the personal sound exposure meter.

A calibrated sound calibrator of a model specified in the instruction manual for the personal sound exposure meter shall be supplied with the personal sound exposure meter. An instruction manual for the sound calibrator and any adaptor required to couple the microphone of the personal sound exposure meter to the sound calibrator shall also be provided. As required by 4.1.5, the model of the sound calibrator shall conform to the relevant specifications of IEC 60942 for the class of sound calibrator.

As appropriate, the laboratory shall utilize the recommendations given in the instruction manual for performing the pattern-evaluation tests.

5.1.2 Display device

All display devices for the personal sound exposure meter shall be verified to be able to display sound levels or sound exposure levels with the resolution required by 4.1.2. The range of the display shall be at least the minimum specified in 4.1.2.

For personal sound exposure meters that can display more than one measurement quantity, it shall be verified that there is a means to ascertain the quantity that is being displayed (see 4.1.2).

If an electrical output is provided on the personal sound exposure meter and the laboratory intends to utilize the electrical output instead of the display device, the laboratory shall verify that changes in the levels of applied acoustical or electrical input signals produce changes in the signal levels indicated on the display device and at the electrical output that are in accordance with the specifications of this document. This requirement applies to each channel of a multi-channel system. Where multiple outputs are present, if an output is specified in the instruction manual for testing, that output should be used for pattern-evaluation tests.

5.1.3 Configuration

All configurations of the personal sound exposure meter that are stated in the instruction manual as conforming to the requirements of this document shall be tested. For all pattern-evaluation tests, the configuration of the personal sound exposure meter, or the multichannel personal sound exposure meter system, shall be as specified in the instruction manual for one of the normal modes of operation, including all required accessories. The configuration shall include a windscreen if a windscreen is an integral component for the normal mode of operation, or if the instruction manual states that the personal sound exposure meter conforms to the specifications of this document with a windscreen installed around the microphone. The model of the windscreen shall be as stated in the instruction manual for use with the personal sound exposure meter.

If the instruction manual states that the personal sound exposure meter conforms to the specifications of this document with optional facilities installed, the configuration with the optional facilities installed shall also be tested to verify conformance to the relevant specifications.

For all tests, the personal sound exposure meter shall be powered from its preferred power supply.

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5.1.4 Design features

5.1.4.1 General

Pattern-evaluation tests described in Clause 5 apply only for those design features that are specified in Clause 4 and that are available in the personal sound exposure meter submitted for test. All such features shall be tested.

If the personal sound exposure meter has more than one signal-processing channel, pattern evaluation tests shall be performed for each channel that utilizes unique signal processing techniques. For multi-channel systems with the same functional equivalence in all channels, the number of channels to be tested may be less than the number of available channels, at the discretion of the laboratory. For a multi-channel system, the number of channels to be tested should be determined from consideration of a scenario for which there is an array of microphones supplying signals to each input with each channel processing the signals in an identical manner. Selection of how many and which channels to test should consider differences, as described in the instruction manual, in the implementation of signal-processing techniques in the various channels. If any special procedure for testing the channels for identical functionality is described in the instruction manual, that procedure should be followed.

NOTE If the personal sound exposure meter is a multi-channel device (for example, a personal sound exposure meter with two or more separate signal inputs with non-parallel processing of digitized data by time-sharing, but quasi-parallel display for the displayed signals), it is usually possible to test the channels for identical functionality either by setting the functions of the channels for identical processing and reading the display(s) or by allowing the channel functionality to rotate by a special test setting procedure thereby allowing comparison of display(s).

5.1.4.2 Mandatory features

A personal sound exposure meter shall be verified to be able to display A-frequency-weighted time-averaged sound level, C-frequency-weighted time-averaged sound level, and C-frequency-weighted peak sound level and to be able to indicate an overload condition.

It shall be verified that a "hold" feature is provided.

If the personal sound exposure meter does not possess the mandatory features listed in 5.1.4.2, the personal sound exposure meter does not conform to the specifications of this document and no pattern-evaluation tests shall be performed.

5.1.4.3 Optional features

If the instruction manual states that any of sound exposure level, sound exposure, percentage criterion sound exposure, percentage criterion non-equal-energy sound exposure or time-weighted sound level are provided, a personal sound exposure meter shall be verified to be able to display the applicable quantities.

5.1.5 Adjustment at the calibration check frequency

Before conducting, but not during, the tests described in Clause 5, the indication of A-weighted sound level at the calibration check frequency shall be checked in accordance with the instruction manual by application of a sound calibrator as specified in 4.1.5. If necessary, the personal sound exposure meter shall be adjusted to indicate the required sound level under reference environmental conditions. For multi-channel devices, the corresponding indications shall be checked for all channels selected for testing.

Environmental conditions at the time of checking the indication shall be recorded.

The effect of environmental conditions on the sound pressure level produced by the sound calibrator shall be accounted for in accordance with the procedure in the instruction manual for the sound calibrator and data from its calibration. The effects shall be evaluated relative to the sound pressure level produced under reference conditions.

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5.1.6 Determination of conformance

No test specified in Clause 5 shall be omitted unless the personal sound exposure meter does not possess the facility described for the test. When the design of a personal sound exposure meter, which has been pattern approved, is changed and a new pattern approval is requested, then, at the discretion of the laboratory, it is not necessary to repeat those tests for electroacoustical performance characteristics that are not affected by the design change.

Conformance to a performance specification is demonstrated when the following criteria are both satisfied:

a) a measured deviation from a design goal does not exceed the applicable acceptance limit;

b) the corresponding uncertainty of measurement does not exceed the corresponding maximum-permitted uncertainty of measurement given in Annex C for the same coverage probability of 95 %.

Annex D gives example assessments of conformance using these criteria.

Laboratories performing pattern evaluation tests shall calculate all uncertainties of measurements in accordance with the guidelines given in ISO/IEC Guide 98-3. Actual measurement uncertainties shall be calculated for a coverage probability of 95 %. Calculation of the actual measurement uncertainty for a particular test should consider at least the following components, as applicable:

– the uncertainty attributed to calibration of the individual instruments and equipment used to perform the test, including the sound calibrator, where applicable;

– the uncertainty resulting from environmental effects or corrections;

– the uncertainty resulting from small errors that may be present in the applied signals;

– the uncertainty attributed to effects associated with the repeatability of the results of the measurements; when a laboratory is only required to perform a single measurement, it is necessary for the laboratory to make an estimate of the random contribution to the total measurement uncertainty; the estimate should be determined from an earlier evaluation of several measurements of the performance of similar personal sound exposure meters;

– the uncertainty associated with the resolution of the display device of the personal sound exposure meter under test; for digital display devices that indicate signal levels with a resolution of 0,1 dB, the uncertainty component should be taken as a rectangular distribution with semi-range of 0,05 dB;

– the uncertainty associated with the device used to mount the personal sound exposure meter in the acoustical test facility;

– the uncertainty resulting from the deviation of the sound field in the acoustical test facility from an ideal sound field;

– the uncertainty associated with each correction applied to the measurement data.

If the uncertainty of measurement exceeds the maximum-permitted uncertainty of measurement, the result of the test shall not be used to demonstrate conformance to a specification and pattern approval shall not be granted.

Tests for conformance to the specifications for the effects of changes in environmental conditions (see 5.3 to 5.8) should preferably be conducted before tests for conformance to the specifications for electroacoustical performance (see 5.9 to 5.26).

5.1.7 Environmental conditions

All tests should be performed at, or referred to, the reference environmental conditions given in 4.1.7.

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The personal sound exposure meter shall be allowed to reach equilibrium with the prevailing environmental conditions before switching on the power to perform a test.

5.1.8 Laboratory equipment

The laboratory shall use instruments with current calibrations for the appropriate quantities. The calibrations shall be traceable to the International System of Units (SI), as required.

5.2 Marking and information in the instruction manual

It shall be verified that the personal sound exposure meter is marked according to the requirements of 4.2.

It shall be verified that the instruction manual contains all the information that is required by 4.27, as relevant to the facilities provided by the personal sound exposure meter.

If the personal sound exposure meter does not conform to the requirements of 5.2, no pattern-evaluation tests shall be performed.

After completion of all tests, the information shall be reviewed to ensure that it is correct and that no applicable acceptance limits are exceeded.

5.3 Static pressure

The influence of static pressure on a personal sound exposure meter shall be tested in accordance with the requirements of IEC 61672-2:2013, 7.1, 7.2, and 7.3, for a class 2 sound level meter.

5.4 Temperature

The influence of temperature on a personal sound exposure meter shall be tested in accordance with the requirements of IEC 61672-2:2013, 7.1, 7.2, 7.4, 7.5, 7.6, and 7.7, for a class 2 sound level meter. The applicable acceptance limits are given in 4.4 of this document.

5.5 Relative humidity

The influence of relative humidity on a personal sound exposure meter shall be tested in accordance with the requirements of IEC 61672-2:2013, 7.1, 7.2, 7.4, 7.5, 7.6, and 7.8, for a class 2 sound level meter. The applicable acceptance limits are given in 4.5 of this document.

5.6 Electrostatic discharge

The influence of electrostatic discharge on a personal sound exposure meter shall be tested in accordance with IEC 61672-2:2013, 7.9.

5.7 Influence of AC-power-frequency and radio-frequency fields

The influence of AC-power-frequency and radio-frequency fields on a personal sound exposure meter shall be tested in accordance with the requirements of IEC 61672-2:2013, 7.10, for a class 2 sound level meter.

5.8 Radio-frequency emissions and public power supply disturbances

Radio-frequency emissions and public power supply disturbances from the personal sound exposure meter shall be tested in accordance with the requirements of IEC 61672-2:2013, Clause 8.

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5.9 Directional response

The directional response of a personal sound exposure meter shall be tested in accordance with the method of IEC 61672-2:2013, 9.1 and 9.3, for a class 2 sound level meter, adapted for the specifications and acceptance limits given in 4.9 of this document.

5.10 Frequency weightings

5.10.1 Acoustical signal tests of frequency weightings

The frequency weightings of a personal sound exposure meter shall be tested by measuring the relative frequency-weighted diffuse-field response using the applicable methods of IEC 61672-2:2013, 9.1 and 9.4, for a class 2 sound level meter.

NOTE For the purposes of this document, relative frequency-weighted diffuse-field response is equivalent to relative frequency-weighted random-incidence response and so the method of IEC 61672-2:2013, 9.4.6, (which requires the use of the free-field method of IEC 61183) is suitable for testing frequency weightings of a personal sound exposure meter.

5.10.2 Electrical signal tests of frequency weightings

The frequency weightings of a personal sound exposure meter shall be tested by measuring the relative frequency-weighted diffuse-field response in accordance with the applicable methods of IEC 61672-2:2013, 9.1 and 9.5, for a class 2 sound level meter.

5.11 Corrections to indicated levels

The corrections to obtain diffuse-field sound levels provided in the instruction manual for a personal sound exposure meter (see 4.11.3) shall be verified in accordance with the method of IEC 61672-2:2013, 9.7, for a class 2 sound level meter.

NOTE For the purposes of this document, diffuse-field sound level is equivalent to random-incidence sound level and so the method of IEC 61672-2:2013, 9.7, is suitable for verifying the corrections for a personal sound exposure meter.

5.12 Linearity of response to steady signals

The linearity of response to steady signals of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-2:2013, 9.8, for a class 2 sound level meter.

5.13 Under-range indication

For each frequency used for level linearity tests, it shall be verified that the indication of an under-range condition is not displayed when the time-weighted sound level, sound exposure level or time-averaged sound level is greater than, or equal to, the lower boundary specified in the instruction manual for the linear operating range. When an under-range condition is indicated, it shall be verified that the under-range indicator operates as specified in 4.13.

5.14 Decay time constants for time weightings F and S

If a personal sound exposure meter provides a means to indicate time-weighted sound level (see 4.1.4.3.6 and 4.14), the personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-2:2013, 9.11.

5.15 Response to short-duration signals for indications of time-averaged sound levels, sound exposure levels, and time-weighted sound levels

Response to short-duration signals of a personal sound exposure meter shall be tested in accordance with the applicable toneburst response tests and repeated toneburst response tests of IEC 61672-2:2013, 9.12, 9.13, and 9.14.

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5.16 Response to short-duration signals for indications of time-averaged non-equal-energy sound levels

If a personal sound exposure meter provides percentage criterion non-equal-energy sound exposure, the following tests of time-average non-equal-energy sound level shall be performed. Responses to 4 kHz sinusoidal electrical signals shall be tested.

Tests of responses to sequences of repeated tonebursts shall begin with a steady 4 kHz sinusoidal electrical signal applied to the personal sound exposure meter. The personal sound exposure meter shall be set initially to indicate A-frequency-weighted time-weighted sound level. The input signal shall be adjusted to give an indication that is 3 dB less than the upper boundary specified in the instruction manual for the linear operating range at 4 kHz. The time-weighted sound level shall be recorded.

Sequences of tonebursts shall be extracted from the steady signal. Single tonebursts in sequences of repeated tonebursts shall have the durations listed in Table 2. Each individual toneburst in a sequence shall begin and end on a zero crossing. The repetition rate for individual tonebursts shall be 10 s. In order to provide stable indications, the total duration of each sequence of tonebursts shall be 5 min. The time-weighted non-equal-energy sound level in response to the sequence of repeated tonebursts shall be recorded for each sequence.

Tests of responses to sequences of repeated tonebursts shall be performed for all applicable individual toneburst durations specified in Table 2 for all provided time weightings and exchange rates.

Measurements of the response to a sequence of repeated tonebursts shall be calculated from the time-averaged non-equal-energy sound level of the sequence minus the corresponding time-weighted sound level of the steady signal.

Deviations of the measured responses to sequences of repeated tonebursts from the corresponding reference toneburst responses shall not exceed the applicable acceptance limits.

5.17 Overload indication

Overload indications of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-2:2013, 9.15.

5.18 C-weighted peak sound level

5.18.1 Deviation from reference differences

Indications of C-weighted peak sound level of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-2:2013, 9.16.

5.18.2 Repeatability

Repeatability of indication of C-weighted peak sound level shall be tested with one-cycle electrical signals as specified in 4.18.2. The one-cycle signals shall begin and end on a zero crossing.

Repeatability of indications of C-weighted peak sound level shall be tested at the levels of three steady input signals from which the one-cycle signals are extracted. The steady input signals shall be those which give indications of C-weighted time-averaged sound level specified in IEC 61672-2:2013, 9.16.

For each signal level, a series of ten measurements of C-weighted peak sound level shall be performed at random intervals during a period of 2 min in response to one-cycle electrical signals with a nominal frequency of 8 kHz.

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For each signal level, the range and standard deviation of the ten indicated C-weighted peak sound levels shall not exceed the specifications given in 4.18.2.

It shall be verified that no overload indication is present for any measurements of C-weighted peak sound level made in accordance with this procedure.

5.19 Stability during continuous operation

Stability during continuous operation of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-3:2013, Clause 15, for class 2 sound level meters.

5.20 High-level stability

High-level stability of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-3:2013, Clause 21, for class 2 sound level meters.

5.21 Reset facility

It shall be verified that operation of the reset facility cancels the previous indication on the display device. It shall also be verified that operation of the reset facility does not give rise to spurious indications on any display device.

5.22 Outputs

The outputs of a personal sound exposure meter shall be tested in accordance with the method of IEC 61672-2:2013, 9.18.

5.23 Timing facilities

The timing facilities of a personal sound exposure meter shall be tested by comparison of the indicated internal averaging time and the actual averaging time. If the personal sound exposure meter provides preset averaging times, the minimum averaging time and the longest preset averaging time that does not exceed 8 h shall be tested. If the personal sound exposure meter does not provide preset averaging times, averaging times of 10 s and 8 h shall be tested.

The relative difference between the indicated internal integration time and the actual integration time shall not exceed the acceptance limits given in 4.23.

The minimum averaging time for measurement of time-averaged sound level and the minimum integration time for measurement of sound exposure level, if provided, shall be verified to be not greater than the corresponding minimum times stated in the instruction manual.

The maximum averaging time and maximum integration time, if applicable, shall be verified to be not less than the corresponding maximum times stated in the instruction manual.

5.24 Crosstalk in multi-channel personal sound exposure meters

Crosstalk between pairs of channels of a multi-channel personal sound exposure meter shall be tested in accordance with the method of IEC 61672-2:2013, 9.20.

5.25 Power supply

The power supply of the personal sound exposure meter shall be tested in accordance with the method of IEC 61672-2:2013, 9.21.

It shall also be verified that the personal sound exposure meter does not continue to record data when the power supply voltage falls below the minimum specified in the instruction manual.

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5.26 Pause and resumption of time averaging and time integration

It shall be confirmed that, while measurements of any time-averaging and time-integrating measurements are paused, the numerical value of all time-averaged or time-integrated quantities for which measurement is in progress shall be retained in memory without modification.

It shall be confirmed that pause or resumption of measurements shall not cause spurious indications on a display device or errors in any stored data.

5.27 Pattern evaluation report

For each personal sound exposure meter that is tested, the pattern evaluation report shall give full details of the configuration that was tested including

– identification of all integral components of the personal sound exposure meter,
– the version of all software that is integral to operation of the personal sound exposure meter,
– the windscreen and accessories that were installed,
– personal sound exposure meter orientations,
– test conditions including environmental conditions, and
– test results.

Each test result shall give the measured deviation from the design goal and the associated actual uncertainty of measurement along with an indication of conformance or non-conformance. A standard format is preferred for reporting the results of pattern evaluation tests.

The test report shall state that the model of the complete personal sound exposure meter conforms to, or does not conform to, the mandatory specifications of this document and hence whether the pattern for the model of the personal sound exposure meter is, or is not, approved. If the model of the personal sound exposure meter is pattern approved, notice of such approval should be made publicly available for use during subsequent periodic tests.

The test information noted in IEC 61000-4-3:2020, Clause 10, shall be included in the test report. The report shall describe any temporary degradation in performance, loss of function, or loss of data noted at the end of a series of tests with electrostatic discharges, AC-power-frequency fields, or radio-frequency fields.

NOTE Although the tests of electromagnetic compatibility given in Clause 5 refer to IEC 61672-2:2013, which refers to IEC 61000-4-3:2006, IEC 61000-4-3:2006/AMD1:2007, and IEC 61000-4-3:2006/AMD2:2010[1], the reporting requirements of IEC 61000-4-3:2020 are compatible with the requirements for testing.

6 Periodic tests

6.1 General requirements

6.1.1 Submission for testing

6.1.1.1 Personal sound exposure meter

An instruction manual applicable to the model and version of the personal sound exposure meter shall be available in order to perform periodic tests of a personal sound exposure meter. If an applicable instruction manual is not submitted along with the personal sound exposure meter, nor available at the laboratory, nor publicly accessible from the Internet web site of the manufacturer or supplier of the personal sound exposure meter, then no periodic tests shall be performed.

[1] This publication, including the amendments, has been withdrawn.

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All items or accessories for the personal sound exposure meter that are necessary for periodic testing shall accompany the personal sound exposure meter when it is submitted for testing. A device specified in the instruction manual for inserting electrical signals equivalent to signals from the microphone shall also be submitted if requested by the laboratory.

Data required to perform the periodic tests shall be available and the source of the data shall be recorded and reported by the laboratory. The data shall include all relevant information required by this document.

6.1.1.2 Sound calibrator

A sound calibrator shall be available. The sound calibrator shall be supplied with the personal sound exposure meter, if required by the user or if requested by the laboratory or shall be supplied by the laboratory. If a suitable sound calibrator is supplied by the user, it shall be used by the laboratory to establish the acoustical sensitivity of the personal sound exposure meter.

The sound calibrator shall preferably be a model specified in the instruction manual of the personal sound exposure meter, or alternatively a sound calibrator producing the same nominal sound pressure level and frequency as a model specified in the instruction manual. In the latter case, any adjustments to be applied when that model of sound calibrator is coupled to the model of microphone supplied on the personal sound exposure meter shall use publicly available data from either the manufacturer of the personal sound exposure meter or the manufacturer of the sound calibrator.

NOTE 1 It is possible that publicly available data for a model of sound calibrator that is not specified in the instruction manual has not been verified according to Clause 5.

Any adaptor required to couple the microphone of the personal sound exposure meter to the sound calibrator shall be available. An instruction manual for the sound calibrator also shall be available. If an appropriate sound calibrator, any necessary adaptor, or the instruction manual are not available, no periodic tests shall be performed.

Evidence shall be available that the sound calibrator conforms to the periodic-testing requirements for generated sound pressure level, frequency, and distortion given for the applicable performance class in IEC 60942. If the evidence of conformance does not include calibrated values of generated sound pressure level, frequency, and distortion, these values shall be available from other evidence such as a suitable calibration certificate.

Evidence of conformance and calibrated values for the sound calibrator may be provided by the user or by the laboratory.

When correction data are used, the source of the data shall be the manufacturer of the sound calibrator, microphone, or personal sound exposure meter, and shall be stated in the documentation.

NOTE 2 There is evidence that reliable measurements of generated sound pressure level can be performed with either a laboratory standard microphone as specified in IEC 61094-1, or a type WS1 microphone as specified in IEC 61094-4, or a type WS2 microphone as specified in IEC 61094-4 with an effective load volume of less than 240 mm^3 and an equivalent volume of less than 12 mm^3.

6.1.2 Preliminary inspection

Prior to any measurements, the personal sound exposure meter and all accessories shall be visually inspected, paying particular attention to damage to, or accumulation of foreign material on, the protection grid or diaphragm of the microphone, if practicable. All relevant controls shall be operated to ensure that they are in working order. It shall be verified that the display device of the personal sound exposure meter is operational.

If the controls, display, and other essential elements are not in proper working order, no periodic tests shall be performed.

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6.1.3 Configuration

For all periodic tests, the configuration of the personal sound exposure meter shall be as requested by the user and shall be as specified in the instruction manual for one of the normal modes of operation, including all required accessories. For a personal sound exposure meter that provides selectable settings for corrections to the relative frequency response for the influence of any accessories or for other configurations of the personal sound exposure meter, the tests shall be performed using the settings that are appropriate for the configuration of the personal sound exposure meter in its normal mode of operation. Any such settings shall remain unchanged throughout the testing.

For a personal sound exposure meter that has an electrical output that is to be used for periodic tests, the difference between the indications obtained from the electrical output and the corresponding indications on the display device of the personal sound exposure meter shall not exceed the specifications given in 4.22. Where multiple outputs are present, if an output is specified in the instruction manual for testing, that output shall be used for periodic testing.

For all tests, the personal sound exposure meter shall be powered from its preferred power supply.

6.1.4 Design features

Periodic tests described in Clause 6 apply only for those design features that are specified in Clause 4 and that are available in the personal sound exposure meter submitted for test. All applicable periodic tests shall be performed.

6.1.5 Adjustment at the calibration check frequency

The indication of the personal sound exposure meter at the calibration check frequency shall be checked by application of the sound calibrator described in 6.1.1.2. The personal sound exposure meter shall be adjusted, if necessary, to indicate the required sound level for the environmental conditions under which the tests are performed. For multi-channel personal sound exposure meter systems, the indication shall be checked for as many of the channels as are required to be tested. The indications of the personal sound exposure meter before and after adjustment shall be recorded. If the adjustment value was determined using a particular model of adaptor with the sound calibrator, an adaptor of the same model shall be used when checking the indication of the personal sound exposure meter.

The effect of the prevailing environmental conditions on the sound pressure level produced in the coupler of the sound calibrator, relative to the sound pressure level produced under the reference environmental conditions of 4.1.7, shall be accounted for in accordance with the procedure and data from the instruction manual for the sound calibrator.

6.1.6 Determination of conformance

Conformance to a performance specification is demonstrated when the following criteria are both satisfied:

a) a measured deviation from a design goal does not exceed the applicable acceptance limit;
b) the corresponding uncertainty of measurement does not exceed the corresponding maximum-permitted uncertainty of measurement given in Annex C for the same coverage probability of 95 %.

Annex D gives example assessments of conformance using these criteria.

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For laboratories performing periodic tests, the uncertainties associated with all measurements shall be determined in accordance with the guidelines given in ISO/IEC Guide 98-3. Metrological terms shall be as defined in ISO/IEC Guide 99. Actual measurement uncertainties shall be calculated for a coverage probability of 95 %. Calculation of the uncertainty of measurement for a particular test should consider at least the following components, as applicable:

– the uncertainty attributed to calibration of the individual instruments and equipment used to perform the test, including the sound calibrator and any free-field facility;
– the uncertainty resulting from environmental effects or adjustments;
– the uncertainty resulting from small errors that may be present in the applied signals;
– the uncertainty attributed to effects associated with the repeatability of the results of the measurements; when a laboratory is only required to make a single measurement, the laboratory shall make an estimate of the contribution of random effects to the total uncertainty; the estimate should be determined from an evaluation of several measurement results previously obtained for a similar personal sound exposure meter;
– the uncertainty associated with the resolution of the display device of the personal sound exposure meter under test; for digital display devices that indicate signal levels with a resolution of 0,1 dB, the uncertainty component should be taken as a rectangular distribution with semi-range of 0,05 dB;
– the uncertainty associated with the device used to mount the personal sound exposure meter in the acoustical test facility;
– the uncertainty resulting from the deviation of the sound field in the acoustical test facility from an ideal sound field;
– the uncertainty associated with each correction applied to the measurement data; if the uncertainties of measurement for the correction data are not available, the uncertainty of the correction data shall be assumed to be the maximum-permitted uncertainty specified in 4.11.1.

Other uncertainty components are described in 6.1.7 and in the applicable test methods.

If an actual uncertainty of measurement for a test performed by the laboratory exceeds the corresponding maximum-permitted uncertainty, the result of the test shall not be used to evaluate conformance to this document for periodic testing.

For tests of frequency weighting (see 6.3), a laboratory's actual uncertainty, without inclusion of the uncertainty of the manufacturer's diffuse-field correction data, shall not exceed the corresponding maximum-permitted uncertainty from Annex C. However, when the uncertainty of the manufacturer's correction data is included, it is possible that an actual uncertainty of measurement may exceed the corresponding maximum-permitted uncertainty only because the uncertainty of the manufacturer's correction data was a significant part of the laboratory's uncertainty budget. In this event, testing may proceed.

6.1.7 Environmental conditions

Periodic tests shall be performed within the following ranges of environmental conditions: 80 kPa to 105 kPa for static air pressure, 20 °C to 26 °C for air temperature, and 25 % to 70 % for relative humidity.

As a minimum, the static air pressure, air temperature, and relative humidity shall be measured and recorded at the start and end of periodic testing.

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Unless data are available for the influence on the model of a personal sound exposure meter of the deviations in the prevailing environmental conditions from the reference environmental conditions, for periodic tests of frequency weightings (see 6.3.1) conducted at a static pressure less than 97 kPa, the laboratory shall assign a standard uncertainty of 0,09 dB for measurements at frequencies less than or equal to 3 kHz and a standard uncertainty of 0,14 dB for measurements at frequencies greater than 3 kHz.

NOTE The two standard uncertainties are for use when values for the standard uncertainties are not available from the instruction manual nor from the Internet website of the manufacturer or supplier of the personal sound exposure meter. The two values were determined from estimates of 0,15 dB and 0,25 dB respectively for the maximum influence of static pressure on frequency response for models of microphones that have the greatest air stiffness in the back cavity.

6.1.8 Laboratory equipment

The laboratory shall use instruments that have been calibrated for the appropriate quantities at appropriate intervals. As required, the calibrations shall be traceable to the SI.

Electrical signals shall be inserted into the personal sound exposure meter through the input device or by the means specified in the instruction manual. The deviation of the frequency of an input signal from a specified frequency shall not exceed ±0,25 % of the specified frequency.

6.1.9 Power supply

Before and after conducting the set of tests with acoustical signals and before and after conducting the set of tests with electrical signals, the power supply for the personal sound exposure meter shall be checked by the method stated in the instruction manual to ensure that it is within the specified operating limits. If the voltage or the equivalent indication of the status of the power supply is not within the operating limits and the reason cannot be attributed to partially discharged batteries or an incorrect selection of the voltage of the public power supply, then no periodic tests shall be performed as a malfunction is indicated.

NOTE Changes to the output of the power supply can be determined as changes to the percentage of the voltage from fully-charged batteries, or by an equivalent procedure.

6.2 Marking

Periodic tests shall not be performed unless the markings on the personal sound exposure meter are as required by 4.2 or there is evidence that the personal sound exposure meter was originally so marked. The serial number and the model designation shall be visible on the personal sound exposure meter.

6.3 Frequency weightings

6.3.1 Acoustical signal tests of a frequency weighting

Frequency weighting C of a personal sound exposure meter shall be tested using acoustical signals by measuring the relative frequency-weighted diffuse-field response at 31,5 Hz, 1 kHz, 4 kHz, and 8 kHz using the applicable methods of IEC 61672-3:2013, Clause 12, for class 2 sound level meters.

NOTE For the purposes of this document, relative frequency-weighted diffuse-field response is equivalent to relative frequency-weighted random-incidence response and so the method of IEC 61672-3:2013, Clause 12, is suitable for testing frequency weightings of a personal sound exposure meter.

6.3.2 Electrical signal tests of frequency weightings

Frequency weightings A and C of a personal sound exposure meter shall be tested using electrical signals by measuring the relative frequency-weighted diffuse-field response at 31,5 Hz, 1 kHz, 4 kHz, and 8 kHz using the applicable methods of IEC 61672-3:2013, Clause 13, for class 2 sound level meters.

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6.4 Frequency and time weightings at 1 kHz

Frequency and time weightings of a personal sound exposure meter at 1 kHz shall be tested in accordance with the applicable methods of IEC 61672-3:2013, Clause 14, for class 2 sound level meters.

6.5 Linearity of response to steady signals

Linearity of response to steady signals of a personal sound exposure meter shall be tested in accordance with the method of IEC 61672-3:2013, Clause 16, for class 2 sound level meters, except that tests shall only be performed for the following signal levels:

– linearity shall be measured from within 5 dB of the specified upper boundary for the linear operating range at 8 kHz then at 1 dB steps of increasing signal level up to, but not including the first indication of overload;
– linearity shall then be measured from within 5 dB of the specified lower boundary, then at 1 dB steps of decreasing input signal level down to the specified lower boundary.

6.6 Response to short-duration signals for indications of time-averaged sound levels

Response to short-duration signals of a personal sound exposure meter shall be tested in accordance with the method of IEC 61672-3, Clause 18, for class 2 sound level meters, except that the tests shall be performed for

– indications of time-averaged sound levels only, and
– tonebursts having duration of 2 ms only.

6.7 Response to short-duration signals for indications of time-averaged non-equal-energy sound levels

If a personal sound exposure meter provides percentage criterion non-equal-energy sound exposure, the response to short-duration signals for indications of time-averaged non-equal-energy sound level shall be tested in accordance with the method of 5.16, except that single tonebursts in sequences of repeated tonebursts shall have durations of 2 ms only.

6.8 Overload indication

Overload indications of a personal sound exposure meter shall be tested in accordance with the method of IEC 61672-3:2013, Clause 20.

6.9 C-weighted peak sound level

6.9.1 Deviation from reference differences

Indications of C-weighted peak sound level of a personal sound exposure meter shall be tested in accordance with the applicable methods of IEC 61672-3:2013, Clause 19, for class 2 sound level meters.

6.9.2 Repeatability

Repeatability of indication of C-weighted peak sound level shall be tested with one-cycle electrical signals using the method described in 5.18.2 at a single signal level. The level of the steady sinusoidal 8 kHz signal from which the single complete cycle is extracted shall yield an indication of C-weighted time-averaged sound level that is 8 dB less that the specified upper boundary of the peak level range.

Application of the one-cycle electrical signal shall not cause indication of an overload condition.

The range and standard deviation of the ten indicated C-weighted peak sound levels shall not exceed the specifications given in 4.18.2.

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6.10 Documentation

The documentation of the periodic test shall contain the information listed in 6.10, as applicable; national regulations can apply.

The documentation of the periodic tests shall contain and shall report prominently one of the following statements regarding conformance or non-conformance:

– when evidence was publicly available to show that pattern-evaluation tests had been performed in accordance with Clause 5 to demonstrate that the model of personal sound exposure meter conformed to all applicable specifications of Clause 4 and the results of all periodic tests according to Clause 6 were satisfactory, a statement as follows:

  "The personal sound exposure meter submitted for testing successfully completed the periodic tests of IEC 61252:2025, for the environmental conditions under which the tests were performed. As evidence was publicly available, from an independent testing organization responsible for approving the results of pattern-evaluation tests performed in accordance with IEC 61252:2025, to demonstrate that the model of personal sound exposure meter fully conformed to the specifications in IEC 61252:2025, the personal sound exposure meter submitted for testing conforms to the specifications of IEC 61252:2025.";

– when no evidence was publicly available to show that pattern-evaluation tests had been performed in accordance with Clause 5 to demonstrate that the model of personal sound exposure meter conformed to all applicable specifications of Clause 4 or if the correction data for acoustical test of frequency weighting were not provided in the instruction manual and the results of all periodic tests according to Clause 6 were satisfactory, a statement as follows:

  "The personal sound exposure meter submitted for testing successfully completed the periodic tests of IEC 61252:2025, for the environmental conditions under which the tests were performed. Because (a) the periodic tests of IEC 61252:2025 cover only a limited subset of the specifications in IEC 61252:2025 and (b) evidence was not publicly available, from an independent testing organization responsible for pattern approvals, to demonstrate that the model of personal sound exposure meter fully conformed to the specifications of IEC 61252:2025 or correction data for acoustical tests of frequency weighting were not provided in the instruction manual, no general statement or conclusion can be made about conformance of the personal sound exposure meter to the full specifications of IEC 61252:2025.";

– when the results of the periodic tests for the personal sound exposure meter were not satisfactory, a statement as follows:

  "The personal sound exposure meter submitted for periodic testing did not successfully complete the tests of IEC 61252:2025. The personal sound exposure meter did not conform to the specifications of IEC 61252:2025.", and a list of which tests were not successfully completed and the reasons therefore.

  EXAMPLE 1 "Measured level linearity deviations exceeded the applicable acceptance limits."

  EXAMPLE 2 "Measured deviations from the design goal for indications of C-weighted peak sound levels exceeded the applicable acceptance limits."

  EXAMPLE 3 "The manufacturer-provided uncertainty for the diffuse-field correction data was a significant part of the laboratory's uncertainty budget and as a result the deviation from the design goal exceeded the acceptance limits for a particular test."

The documentation of the periodic test shall also contain the following:

a) the date(s) when the periodic tests were performed;

b) the statement: "Periodic tests were performed in accordance with procedures from IEC 61252:2025";

c) a statement of the availability or non-availability of (and, if available, a reference to) evidence, from an independent testing organization responsible for pattern approvals, to demonstrate that the model of personal sound exposure meter submitted for periodic testing successfully completed the applicable pattern-evaluation tests given in Clause 5;

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d) the name and location of the laboratory performing the periodic tests;

e) the name of the manufacturer or supplier, model designation, and serial number of the personal sound exposure meter and, if applicable, the version of all software that is integral to operation of the personal sound exposure meter;

f) the name of the manufacturer or supplier, model designation and serial number of the microphone and any components that are separable from the personal sound exposure meter;

g) if the personal sound exposure meter is a multi-channel device, a designation of which channels were selected for testing;

h) an identification of the instruction manual of the personal sound exposure meter including, as applicable, the publication date and version number; for instruction manuals downloaded from an Internet website, the date of the download as well as any unique descriptive information;

i) the name of the manufacturer or supplier, model designation and serial number of the sound calibrator, together with details of any adaptors used and the source of any correction data used to adjust the indication at the calibration check frequency; if applicable, a statement that the sound calibrator is not of a model that is specified in the instruction manual of the personal sound exposure meter;

j) a description of the source of the data used to correct the sound levels indicated in response to application of a multi-frequency sound calibrator, comparison coupler, or electrostatic actuator to sound levels equivalent to those that would be indicated in response to a diffuse sound field; if applicable, a statement that the source of the correction data is not the instruction manual of the personal sound exposure meter; if applicable, a statement of evidence of validation of the correction data;

k) when the uncertainties of measurement for the diffuse-field correction data are not available (see 6.1.6), a statement as follows: "No information on the uncertainty of measurement, required by IEC 61252:2025, for the correction data given in the instruction manual or obtained from the manufacturer or supplier of the personal sound exposure meter, or the manufacturer of the microphone, or the manufacturer of the multi-frequency sound calibrator, comparison coupler, or electrostatic actuator (delete as appropriate) was provided in the instruction manual or made available by the manufacturer or supplier of the personal sound exposure meter. The uncertainty of measurement of the correction data was therefore assumed to be the maximum-permitted uncertainty specified by IEC 61252:2025 for the applicable correction data."

l) if an actual uncertainty of measurement exceeds the corresponding maximum-permitted uncertainty only because the uncertainty of the manufacturer's correction data was a significant part of the laboratory's uncertainty budget (see 6.1.6), a statement to explain the reasons that the results of a test did not conform to a specification of Clause 4; the statement should include a reference to the applicable requirement of this document (see 6.1.6);

m) a statement of the calibration check frequency, reference sound pressure level, and reference level range for the personal sound exposure meter;

n) a description of the configuration of the personal sound exposure meter for the tests including any connecting cables that were provided to operate the personal sound exposure meter;

o) the ranges of the static pressure, air temperature, and relative humidity measured during the testing;

p) from application of the calibrated sound calibrator, the initial and adjusted indications of the personal sound exposure meter at the calibration check frequency, and reference to the certificate for the sound calibrator, if available;

q) for each test result, the measured deviation from the design goal along with the associated maximum-permitted uncertainty of measurement and the laboratory's actual uncertainty for each measurement result, unless it is agreed to provide these results in a simplified way.

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Annex A
(informative)

Sound exposures and corresponding normalized 8 h-average sound levels

Table A.1 lists example values of sound exposure and corresponding normalized 8 h-average sound level.

Table A.1 – Sound exposures and corresponding normalized 8 h-average sound levels

Sound exposure Pa^2h	Normalized 8 h-average sound level dB
0,32	80
0,40	81
0,51	82
0,64	83
0,80	84
1,01	85
1,27	86
1,60	87
2,02	88
2,54	89
3,20	90
4,03	91
5,07	92
6,38	93
8,04	94
10,12	95
12,74	96
16,04	97
20,19	98
25,42	99
32,00	100
40,29	101
50,72	102
63,85	103
80,38	104
101,19	105
NOTE For normalized 8 h-average sound levels, $L_{Aeq,8\ hn}$, in decibels, sound exposures E, in pascal-squared hours, were calculated from Formula (3) for p_0 = 20 µPa and T_n = 8 h.	

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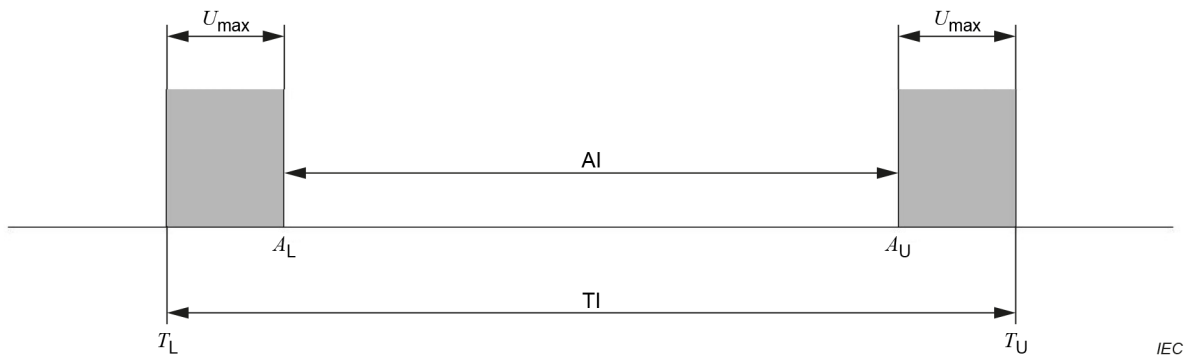
Annex B
(informative)

Relationship between tolerance interval, corresponding acceptance interval and the maximum-permitted uncertainty of measurement

This document, in common with others written by IEC/TC 29, uses adaptations of the guidelines from ISO/IEC Guide 98-4 (equivalent to guidance document JCGM 106 from the Joint Committee for Guides in Metrology), as the basis for demonstration of conformance of an instrument to the specifications given in this document.

ISO/IEC Guide 98-4 describes guarded acceptance in terms of tolerance intervals, acceptance intervals and uncertainties of measurement.

To promote clarity for users and testing laboratories, IEC/TC 29 has adopted a policy whereby tolerance limits around design goals are not explicitly stated, but can be determined if required from the specified acceptance limits for allowed deviations from a design goal and the corresponding specified maximum-permitted uncertainty of measurement, by using the illustration in Figure B.1.



Key

AI acceptance interval

TI tolerance interval

U_{max} guard band for the maximum-permitted uncertainty of measurement for a 95 % coverage interval

A_L lower acceptance limit

A_U upper acceptance limit

T_L lower tolerance limit

T_U upper tolerance limit

Figure B.1 – Relationship between tolerance interval, corresponding acceptance interval and the maximum-permitted uncertainty of measurement

The limits of an acceptance interval are associated with the acceptance interval and not with the guard band for the maximum-permitted uncertainty of measurement. Hence a measured deviation equal to a limit of an acceptance interval demonstrates conformance to a specification, providing also that the uncertainty of the measurement from the laboratory performing a test does not exceed the specified maximum-permitted uncertainty.

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Annex C
(normative)

Maximum-permitted uncertainties of measurement

For the requirements for which the test methods for pattern-evaluation tests given in IEC 61672-2 and periodic tests given in IEC 61672-3 are used to determine conformance to the applicable specifications of this document, the maximum-permitted uncertainties are given in IEC 61672-1:2013, Annex B. Those requirements (with the applicable subclauses of this document in parentheses) are:

– static pressure influence (see 4.3),
– temperature influence (see 4.4),
– relative humidity influence (see 4.5),
– combined temperature and relative humidity influence (see 4.4 and 4.5),
– AC-power-frequency and radio-frequency fields (see 4.7),
– directional response (see 4.9),
– frequency weightings (see 4.10),
– linearity of response to steady signals (see 4.12),
– time-weightings F and S (see 4.14),
– response to short-duration signals for indications of time-averaged sound levels, sound exposure levels, and time-weighted sound levels (see 4.15),
– overload indication (see 4.17),
– C-weighted peak sound level (see 4.18),
– stability during continuous operation (see 4.19),
– high-level stability (see 4.20),
– analogue output (see 4.22),
– power supply (see 4.25).

For the response to short-duration signals for indications of time-averaged non-equal-energy sound levels and for timing facilities, Table C.1 gives the maximum-permitted uncertainties, for a coverage probability of 95 %, applicable to pattern-evaluation tests and periodic tests to demonstrate conformance of a personal sound exposure meter to the specifications of this document.

NOTE The maximum-permitted uncertainties of measurement in IEC 61672-1:2013, Annex B, or in Table C.1 of this document are not equivalent to the uncertainties associated with the measurement of a sound level.

Table C.1 – Maximum-permitted uncertainties of measurement for a coverage probability of 95 % for the response to short-duration signals for indications of time-averaged non-equal-energy sound levels and for timing facilities

Requirement	Subclause	Maximum-permitted uncertainty of measurement
Response to short-duration signals for indications of time-averaged non-equal-energy sound levels	4.16	0,40 dB
Timing facilities	4.23	0,1 %

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Annex D
(informative)

Example assessments of conformance to specifications of this document

D.1 General

The purpose of Annex D is to clarify the use of measurement results and uncertainties of measurement in assessments of conformance to the specifications of Clause 4 in either pattern-evaluation tests (see Clause 5) or periodic tests (see Clause 6) of personal sound exposure meters.

Annex D demonstrates assessment of conformance using some general illustrative examples.

D.2 Conformance criteria

According to the requirements in this document, conformance to a specification is established when measured deviations from design goals do not exceed the corresponding acceptance limits and the uncertainty of measurement does not exceed the corresponding maximum-permitted uncertainty of measurement for a coverage probability of 95%.

With these two criteria, there are four possible outcomes:

1) measured deviations do not exceed acceptance limits and actual uncertainty does not exceed maximum-permitted uncertainty: conformance to the specification;
2) measured deviations do not exceed acceptance limits and actual uncertainty exceeds maximum-permitted uncertainty: non-conformance because the actual uncertainty exceeds the maximum-permitted uncertainty;
3) measured deviations exceed acceptance limits and actual uncertainty does not exceed maximum-permitted uncertainty: non-conformance because measured deviations exceed the acceptance limits;
4) measured deviations exceed acceptance limits and actual uncertainty exceeds maximum-permitted uncertainty: non-conformance because neither criterion is satisfied.

NOTE In practice, a laboratory can sometimes pre-determine the uncertainty of a measurement. If the predetermined uncertainty exceeds the maximum-permitted uncertainty, the laboratory would not attempt to perform the test.

D.3 Example test results

Table D.1 gives examples of test results to explain the method of determining conformance or non-conformance to the specifications of this document. This method applies for any tests in this document where acceptance limits and maximum-permitted uncertainties are specified.

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Table D.1 – Examples of assessment of conformance

Example number	Measured deviation from design goal dB	Acceptance limits dB	Actual uncertainty dB	Maximum–permitted uncertainty dB	Conforms to specifications Yes or no	Reasons for conformance or non-conformance
1	+1,7	+1,0 −1,2	0,3	0,5	No	Deviation exceeds acceptance limits
2	+1,1	+1,0 −1,2	0,3	0,5	No	Deviation exceeds acceptance limits
3	+1,0	+1,0 −1,2	0,3	0,5	Yes	Deviation does not exceed acceptance limits and uncertainty within maximum-permitted
4	0,0	+1,0 −1,2	0,3	0,5	Yes	Deviation does not exceed acceptance limits and uncertainty within maximum-permitted
5	0,0	+1,0 −1,2	0,9	0,5	No	Deviation does not exceed acceptance limits but uncertainty exceeds maximum-permitted
6	−0,5	+1,0 −1,2	0,3	0,5	Yes	Deviation does not exceed acceptance limits and uncertainty within maximum-permitted
7	−1,2	+1,0 −1,2	0,3	0,5	Yes	Deviation does not exceed acceptance limits and uncertainty within maximum-permitted
8	−1,3	+1,0 −1,2	0,3	0,5	No	Deviation exceeds acceptance limits
9	−2,0	+1,0 −1,2	0,3	0,5	No	Deviation exceeds acceptance limits
10	−2,0	+1,0 −1,2	0,7	0,5	No	Deviation exceeds acceptance limits and uncertainty exceeds maximum-permitted

Figure D.1 shows the ten example assessments of conformance from Table D.1 in graphical form.

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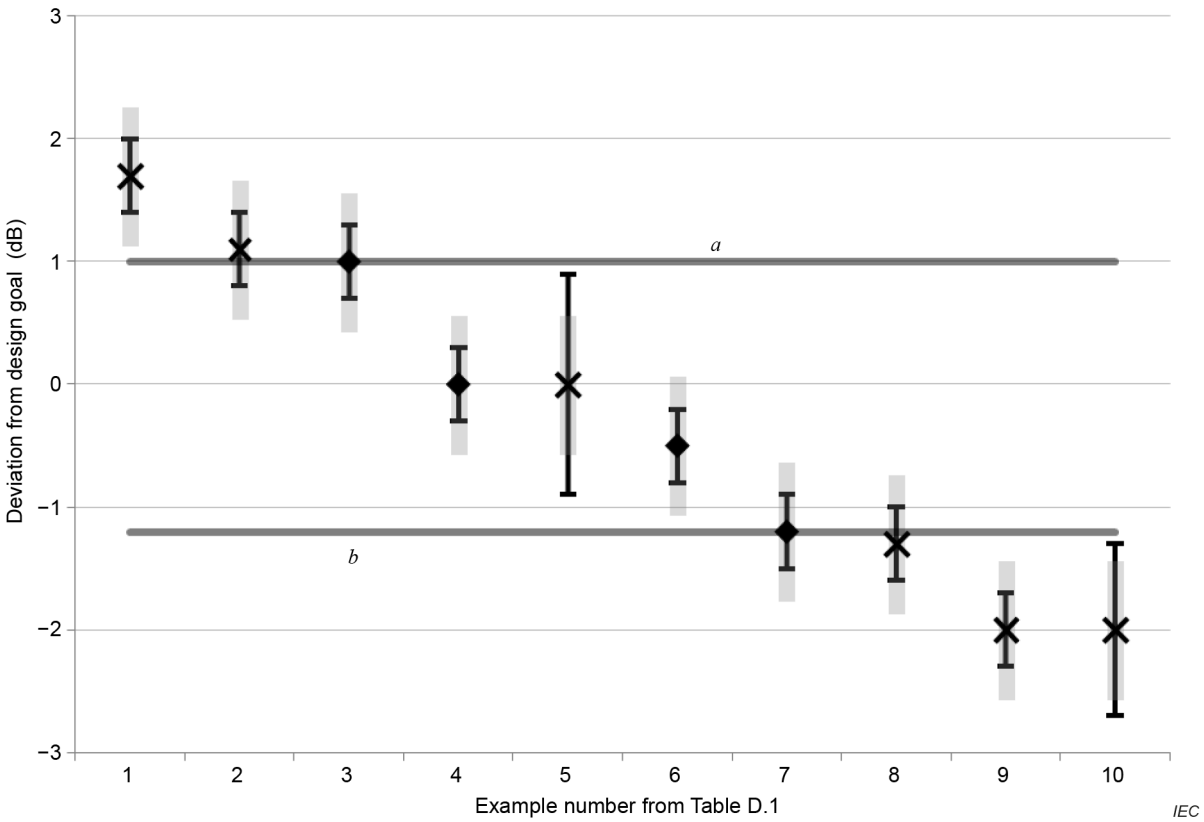

Key

a upper acceptance limit

b lower acceptance limit

A diamond-shaped marker indicates conformance to the specification and a cross-shaped marker indicates non-conformance. The actual uncertainty of measurement is indicated by the vertical error bars and the maximum-permitted uncertainty is indicated by the vertical shaded area.

Figure D.1 – Examples of assessment of conformance

In Figure D.1, the lower and upper acceptance limits are indicated by the heavy horizontal lines. The measured deviations from the design goal are shown by the solid markers. A diamond-shaped marker indicates conformance to the specification and a cross-shaped marker indicates non-conformance.

In Figure D.1, the actual uncertainty of measurement is indicated by the vertical error bars and the maximum-permitted uncertainty is indicated by the vertical shaded area.

The practice illustrated in Table D.1 and Figure D.1 for assessing conformance applies equally for pattern-evaluation testing as well as periodic testing.

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[2] This publication, including the amendments, has been withdrawn.

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